Chapter 9: Phase Diagrams

ISSUES TO ADDRESS...

- When we combine two elements...

  what equilibrium state do we get?

- In particular, if we specify...

  --a composition (e.g., wt% Cu - wt% Ni), and

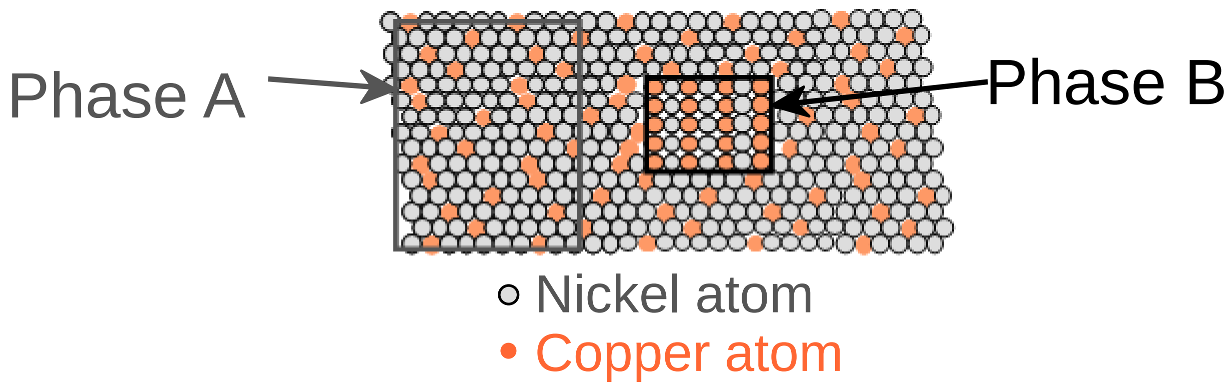
  --a temperature (T)

then...

How many phases do we get?

What is the composition of each phase?

How much of each phase do we get?



Phase Equilibria: Solubility Limit

Introduction

- Solutions – solid solutions, single phase
- Mixtures – more than one phase

Adapted from Fig. 9.1, Callister 7e.

- Solubility Limit:
  Max concentration for which only a single phase solution occurs.

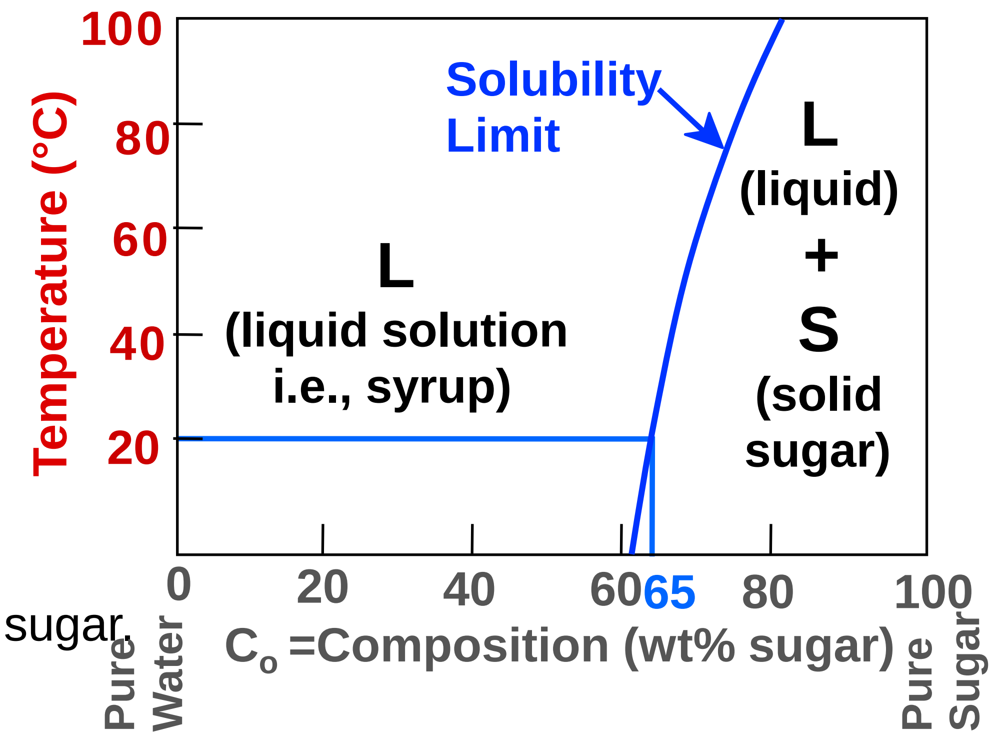
Question: What is the solubility limit at 20°C?

Answer: 65 wt% sugar.

If C_o < 65 wt% sugar: syrup

If C_o > 65 wt% sugar: syrup + sugar.

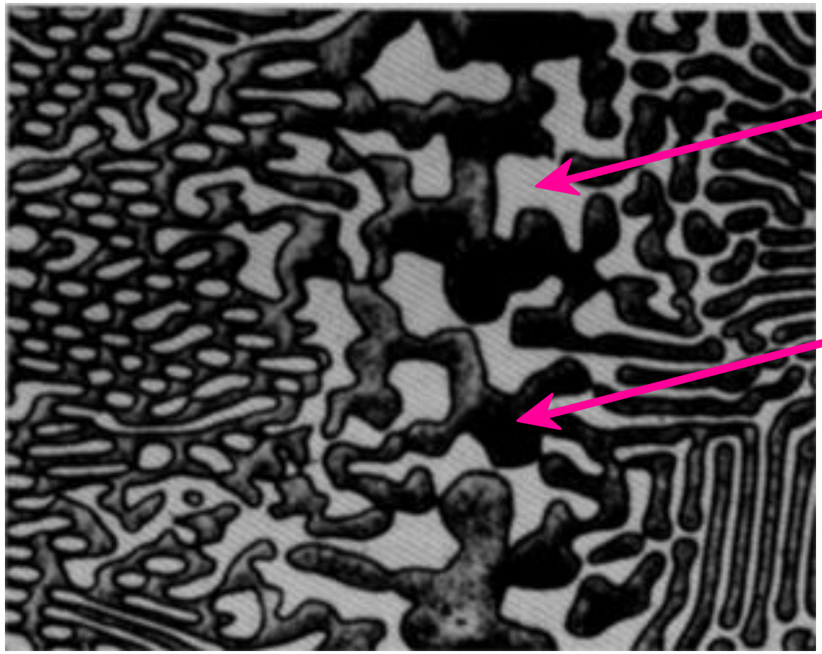
Sucrose/Water Phase Diagram



Components and Phases

- Components:
  The elements or compounds which are present in the mixture (e.g., Al and Cu)
- Phases:
  The physically and chemically distinct material regions that result (e.g., α and β).

Aluminum-Copper Alloy



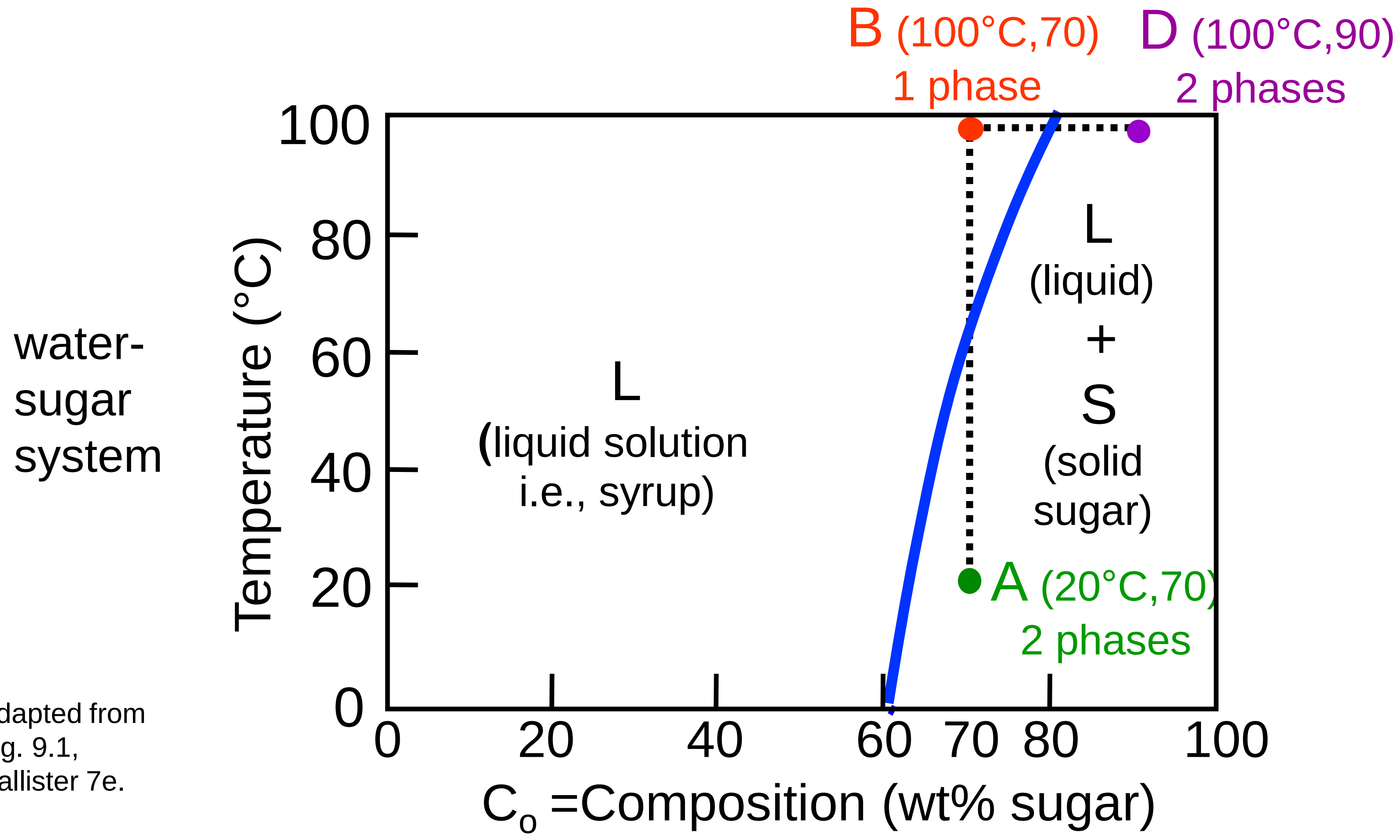
β (lighter phase)

α (darker phase)

Adapted from chapter-opening photograph, Chapter 9, Callister 3e.

Effect of *T* & Composition (C_o)

- Changing T can change # of phases: path A to B.
- Changing C_o can change # of phases: path B to D.



Adapted from
Fig. 9.1,
Callister 7e.

Phase Equilibria

Simple solution system (e.g., Ni-Cu solution)

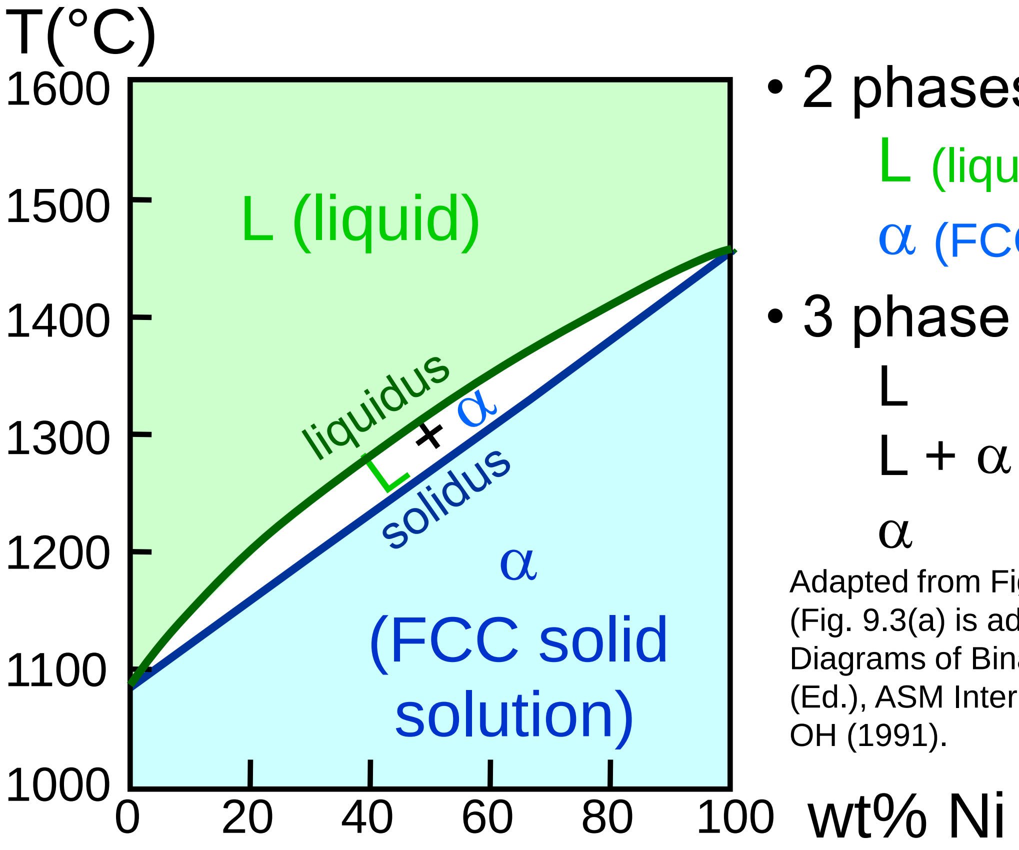
	Crystal Structure	electroneg	*r* (nm)
Ni	FCC	1.9	0.1246
Cu	FCC	1.8	0.1278

- Both have the same crystal structure (FCC) and have similar electronegativities and atomic radii (W. Hume – Rothery rules) suggesting high mutual solubility.
- Ni and Cu are totally miscible in all proportions.

Phase Diagrams

- Indicate phases as function of T, C_o, and P.
- For this course:
  - -binary systems: just 2 components.
  - -independent variables: T and C_o (P = 1 atm is almost always used).

- Phase Diagram for Cu-Ni system



- 2 phases:
  - L (liquid)
  - α (FCC solid solution)
- 3 phase fields:
  - L
  - L + α
  - α

Adapted from Fig. 9.3(a), Callister 7e. (Fig. 9.3(a) is adapted from Phase Diagrams of Binary Nickel Alloys, P. Nash (Ed.), ASM International, Materials Park, OH (1991).

Phase Diagrams:

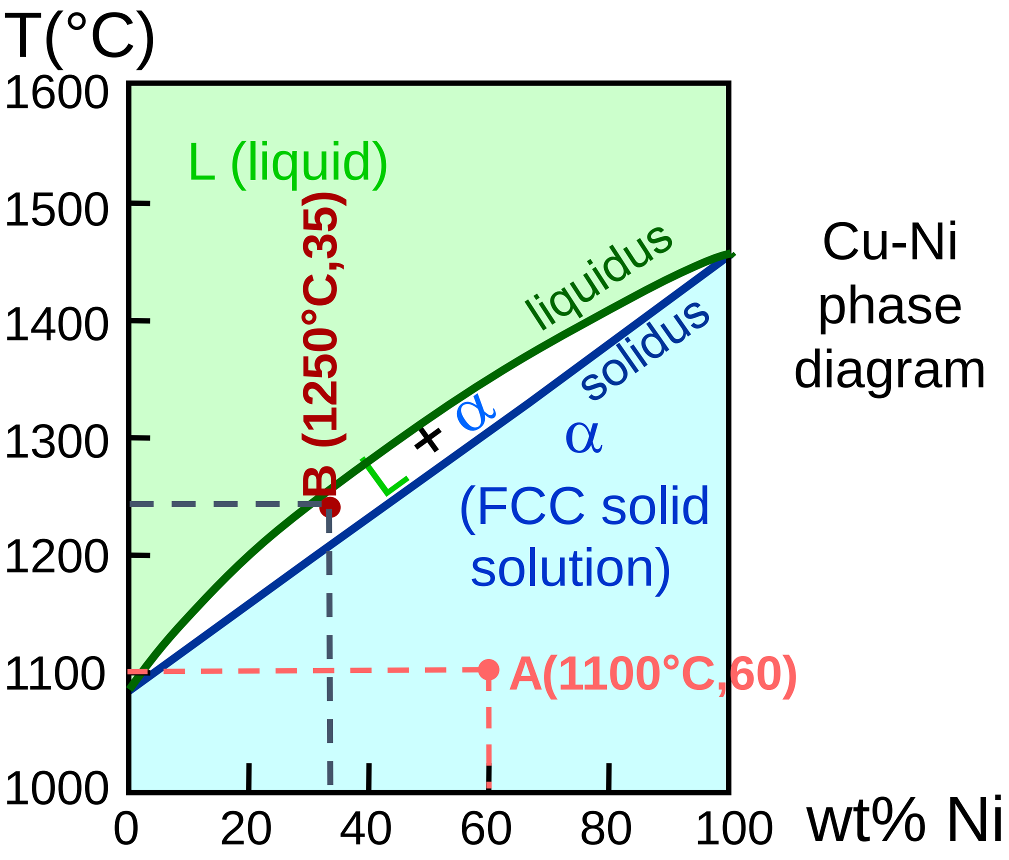
and types of phases

- Rule 1: If we know T and C_o, then we know:
  --the # and types of phases present.

- Examples:

  A(1100°C, 60):
  1 phase: α

  B(1250°C, 35):
  2 phases: L + α



Cu-Ni phase diagram

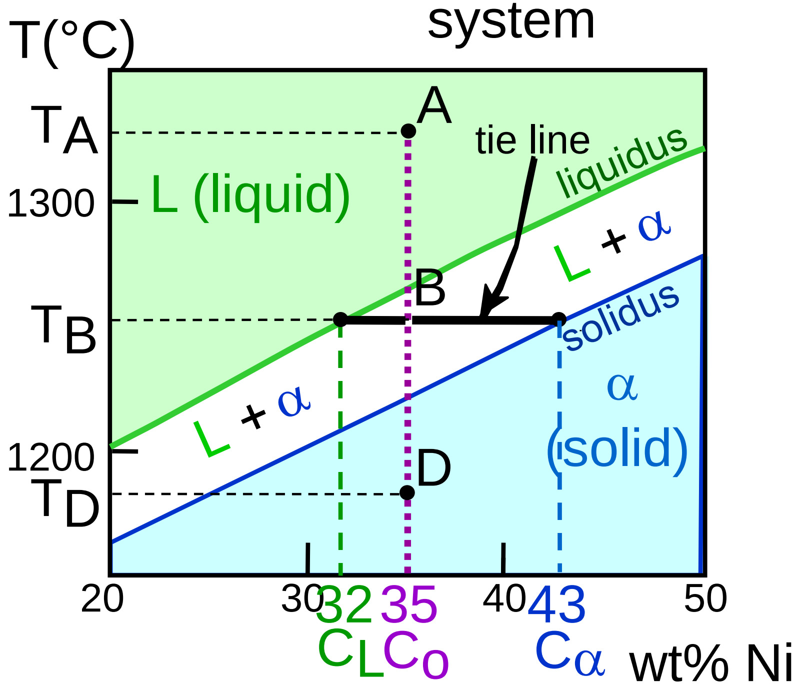
Adapted from Fig. 9.3(a), Callister 7e. (Fig. 9.3(a) is adapted from Phase Diagrams of Binary Nickel Alloys, P. Nash (Ed.), ASM International, Materials Park, OH, 1991).

Phase Diagrams:
composition of phases

- Rule 2: If we know T and C_o, then we know:
  --the composition of each phase.

- Examples:
  C_o = 35 wt% Ni
  At T_A = 1320°C:
  Only Liquid (L)
  $C_L = C_o$ (= 35 wt% Ni)
  At T_D = 1190°C:
  Only Solid (α)
  $C_\alpha = C_o$ (= 35 wt% Ni)
  At T_B = 1250°C:
  Both α and L
  $C_L = C_{liquidus}$ (= 32 wt% Ni here)
  $C_\alpha = C_{solidus}$ (= 43 wt% Ni here)

Cu-Ni system



Adapted from Fig. 9.3(b), Callister 7e. (Fig. 9.3(b) is adapted from Phase Diagrams of Binary Nickel Alloys, P. Nash (Ed.), ASM International, Materials Park, OH, 1991.)

Phase Diagrams:
weight fractions of phases

- Rule 3: If we know T and C_o, then we know:
  --the amount of each phase (given in wt%).
- Examples:

C_o = 35 wt% Ni

At T_A: Only Liquid (L)

W_L = 100 wt%, W_α = 0

At T_D: Only Solid (α)

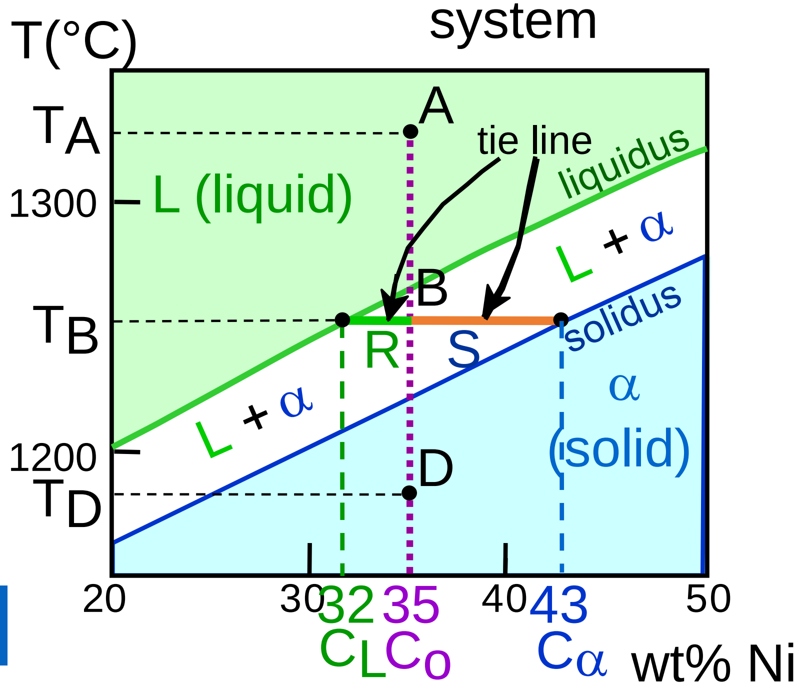
W_L = 0, W_α = 100 wt%

At T_B: Both α and L

$$W_L = \frac{S}{R+S} = \frac{43-35}{43-32} = 73\ wt\ \%$$

$$W_\alpha = \frac{R}{R+S} = 27\ wt\%$$

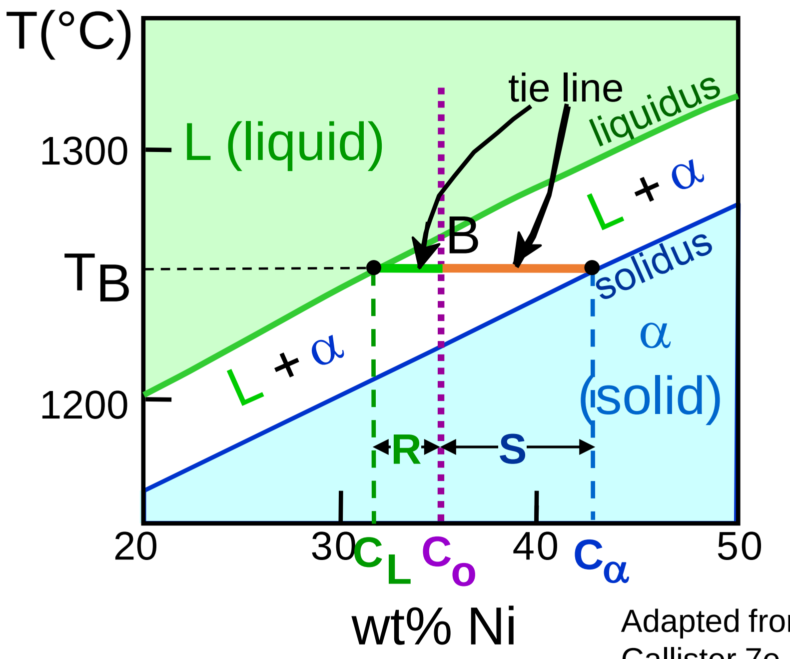
Cu-Ni system



Adapted from Fig. 9.3(b), Callister 7e.
(Fig. 9.3(b) is adapted from Phase Diagrams of Binary Nickel Alloys, P. Nash (Ed.), ASM International, Materials Park, OH, 1991.)

The Lever Rule

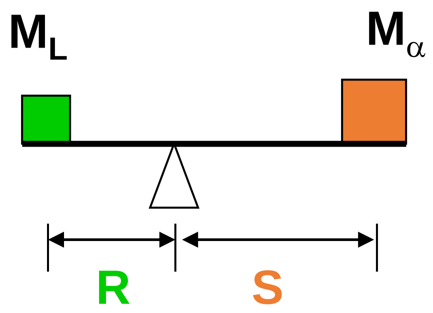
- Tie line – connects the phases in equilibrium with each other - essentially an isotherm



Adapted from Fig. 9.3(b), Callister 7e.

How much of each phase?

Think of it as a lever (teeter-totter)



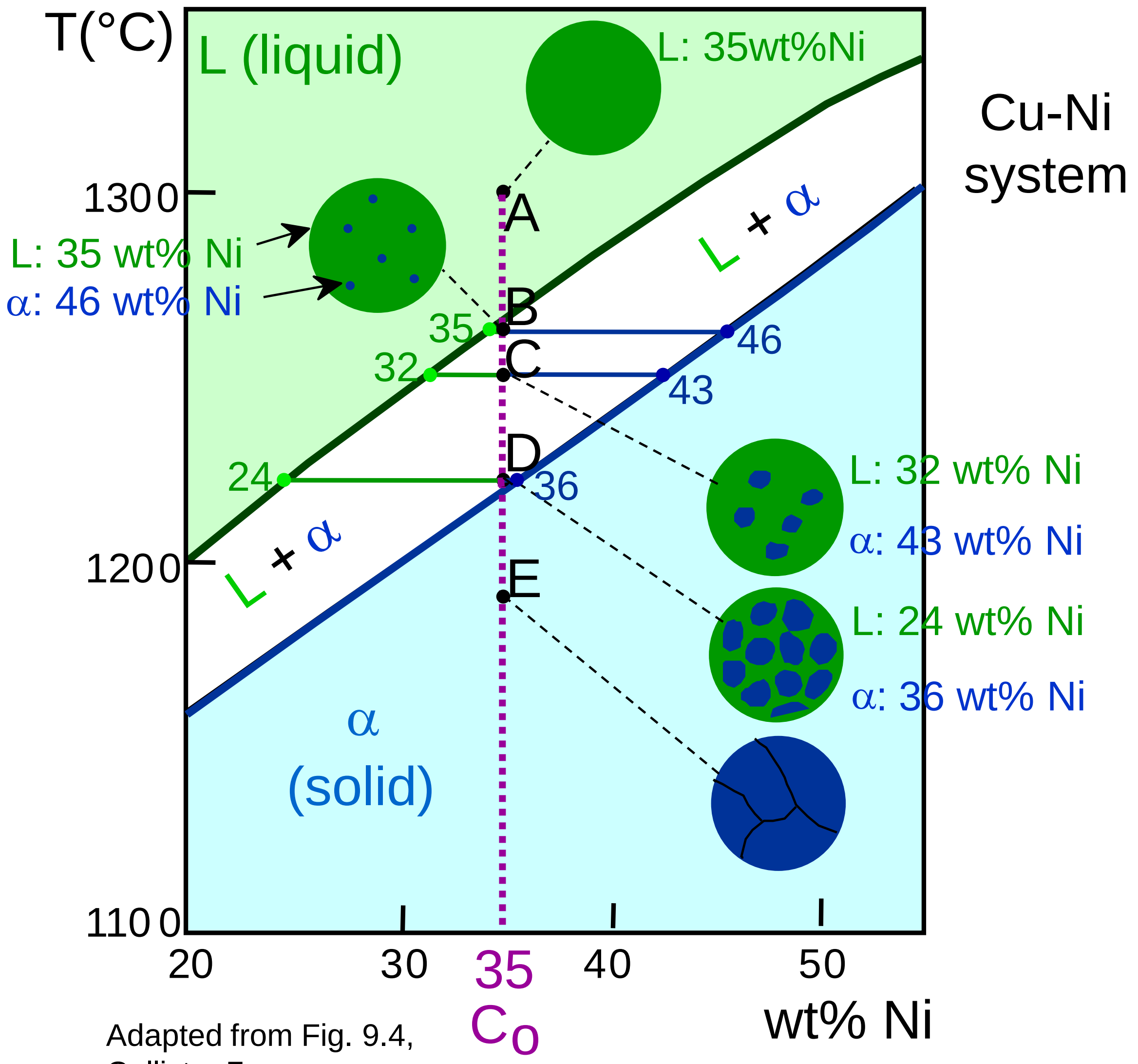
$$M_\alpha \cdot S = M_L \cdot R$$

$$W_L = \frac{M_L}{M_L + M_\alpha} = \frac{S}{R+S} = \frac{C_\alpha - C_0}{C_\alpha - C_L}$$

$$W_\alpha = \frac{R}{R+S} = \frac{C_0 - C_L}{C_\alpha - C_L}$$

Ex: Cooling in a Cu-Ni Binary

- Phase diagram: Cu-Ni system.
- System is:
  --binary i.e., 2 components: Cu and Ni.
  --isomorphous i.e., complete solubility of one component in another; α phase field extends from 0 to 100 wt% Ni.
- Consider C_o = 35 wt%Ni.



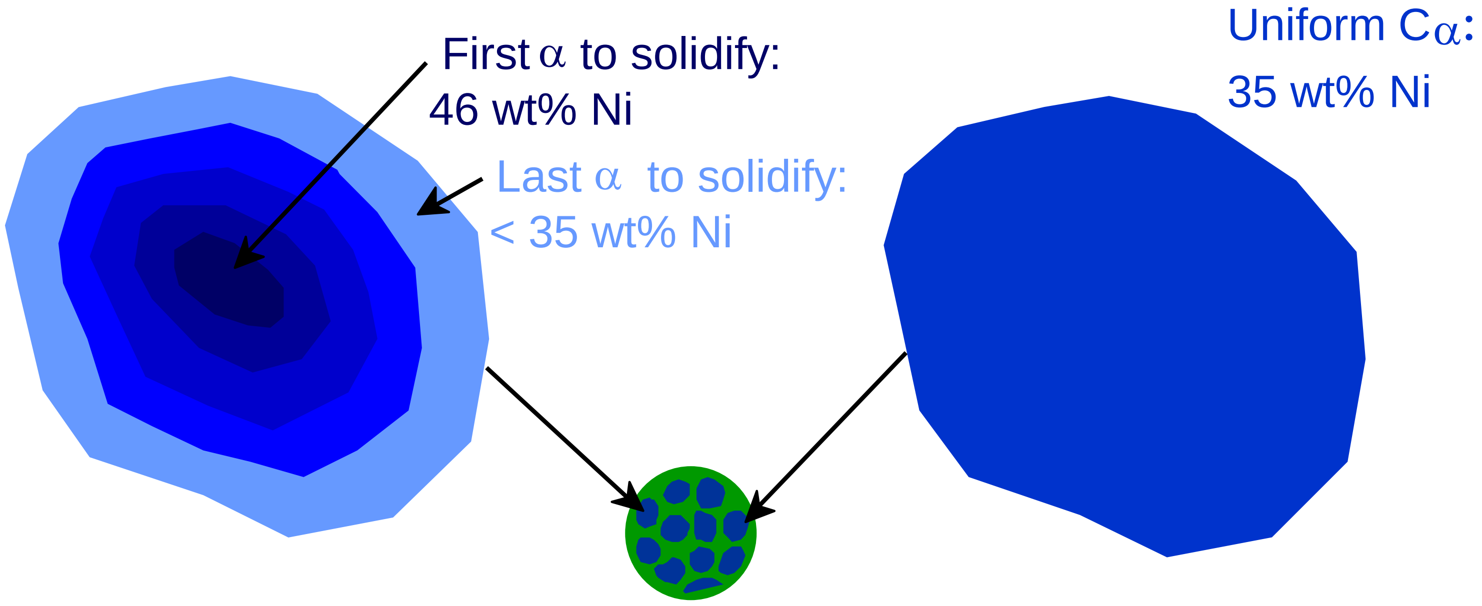
Adapted from Fig. 9.4, Callister 7e.

Cored vs Equilibrium Phases

- C_α changes as we solidify.
- Cu-Ni case: First α to solidify has C_α = 46 wt% Ni.
  Last α to solidify has C_α = 35 wt% Ni.

- Fast rate of cooling:
  Cored structure

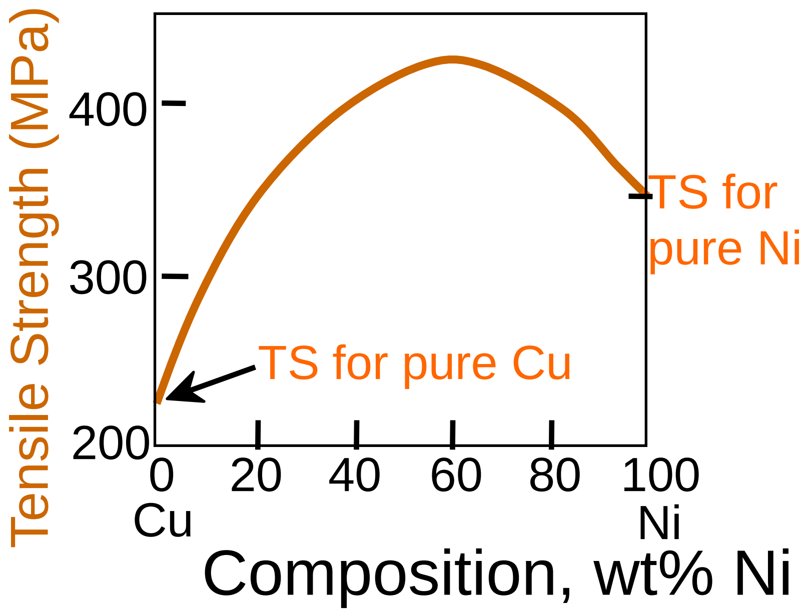
- Slow rate of cooling:
  Equilibrium structure



Mechanical Properties: Cu-Ni System

- Effect of solid solution strengthening on:

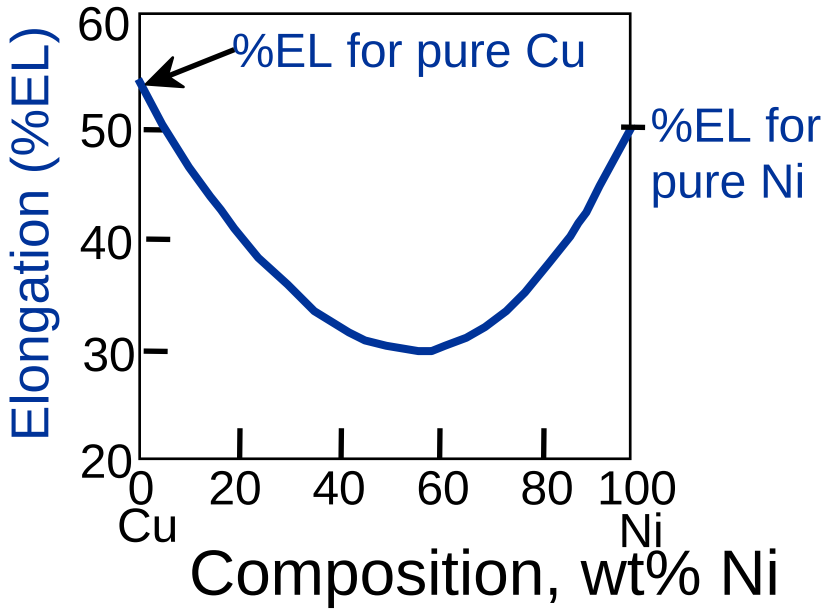
--Tensile strength (TS)



Adapted from Fig. 9.6(a), Callister 7e.

--Peak as a function of C_o

--Ductility (%EL,%AR)



Adapted from Fig. 9.6(b), Callister 7e.

--Min. as a function of C_o

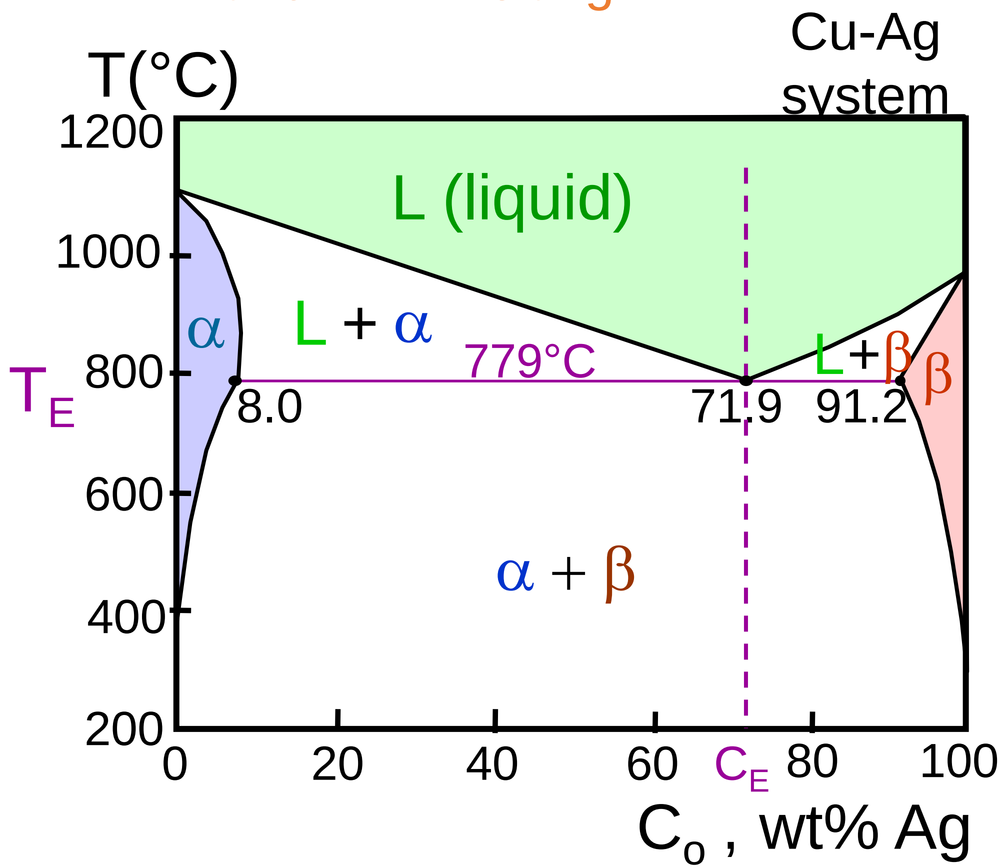
Binary-Eutectic Systems

2 components

has a special composition
with a min. melting T.

Ex.: Cu-Ag system

- 3 single phase regions
  (L, α, β)
- Limited solubility:
  α: mostly Cu
  β: mostly Ag
- T_E : No liquid below T_E
- C_E : Min. melting T_E
  composition



Adapted from Fig. 9.7, Callister 7e.

- **Eutectic transition**

$$L(C_E) \rightleftharpoons \alpha(C_{\alpha E}) + \beta(C_{\beta E})$$

EX: Pb-Sn Eutectic System (1)

- For a 40 wt% Sn-60 wt% Pb alloy at 150°C, find...

--the phases present: α + β

--compositions of phases:

C_O = 40 wt% Sn

C_α = 11 wt% Sn

C_β = 99 wt% Sn

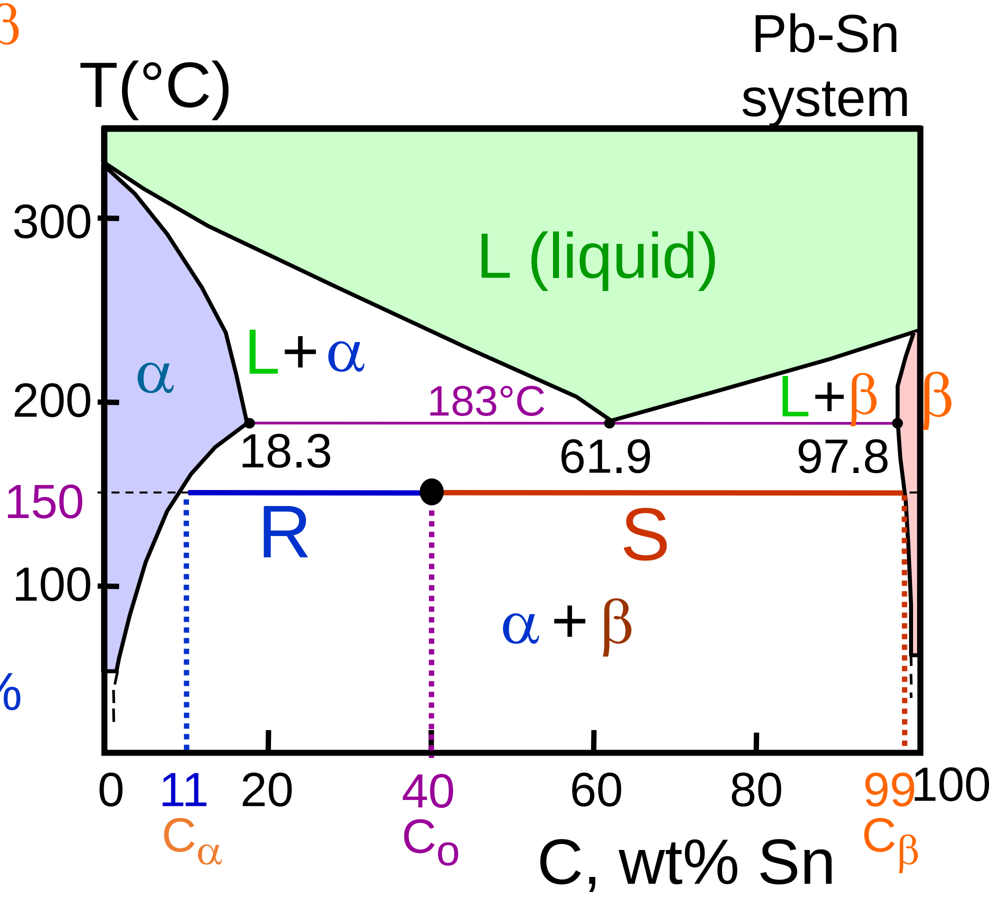
--the relative amount of each phase:

$$W_\alpha = \frac{S}{R+S} = \frac{C_\beta - C_O}{C_\beta - C_\alpha}$$

$$= \frac{99 - 40}{99 - 11} = \frac{59}{88} = 67 \text{ wt\%}$$

$$W_\beta = \frac{R}{R+S} = \frac{C_O - C_\alpha}{C_\beta - C_\alpha}$$

$$= \frac{40 - 11}{99 - 11} = \frac{29}{88} = 33 \text{ wt\%}$$



Adapted from Fig. 9.8, Callister 7e.

EX: Pb-Sn Eutectic System (2)

- For a 40 wt% Sn-60 wt% Pb alloy at 220°C, find...

--the phases present: α + L

--compositions of phases:

C_O = 40 wt% Sn

C_α = 17 wt% Sn

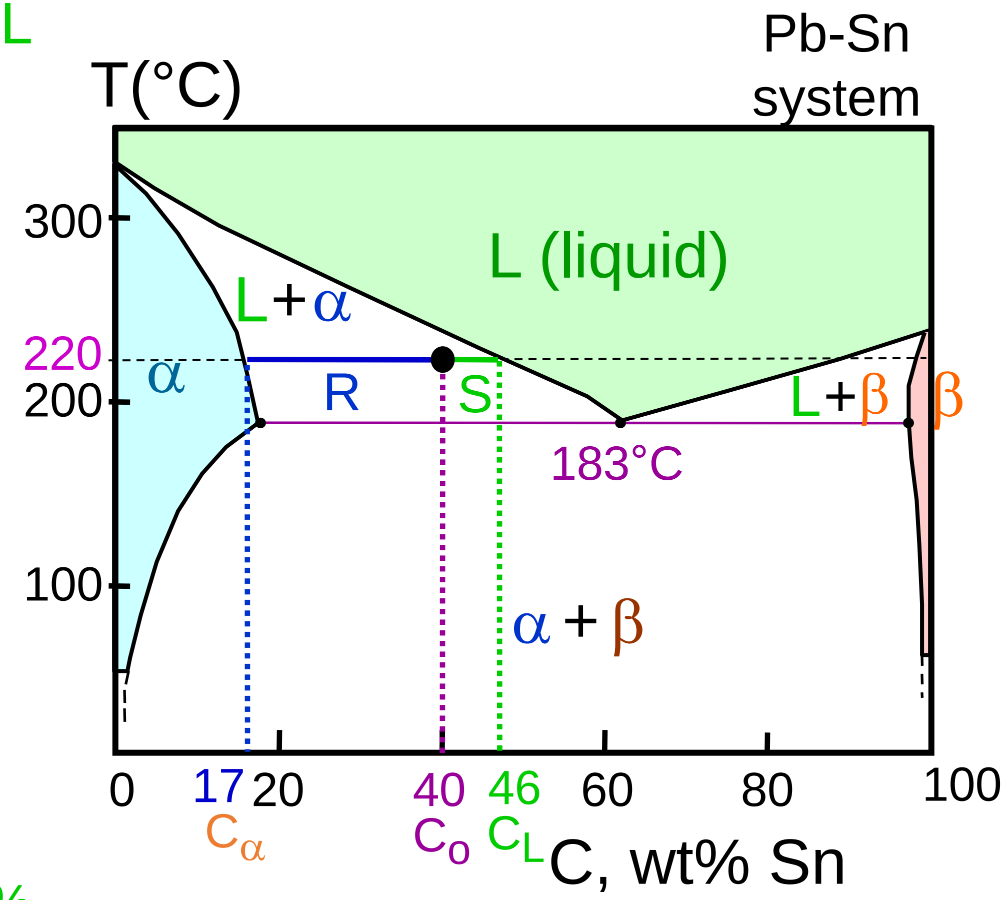
C_L = 46 wt% Sn

--the relative amount of each phase:

$$W_\alpha = \frac{C_L - C_O}{C_L - C_\alpha} = \frac{46 - 40}{46 - 17}$$

$$= \frac{6}{29} = 21 \text{ wt\%}$$

$$W_L = \frac{C_O - C_\alpha}{C_L - C_\alpha} = \frac{23}{29} = 79 \text{ wt\%}$$



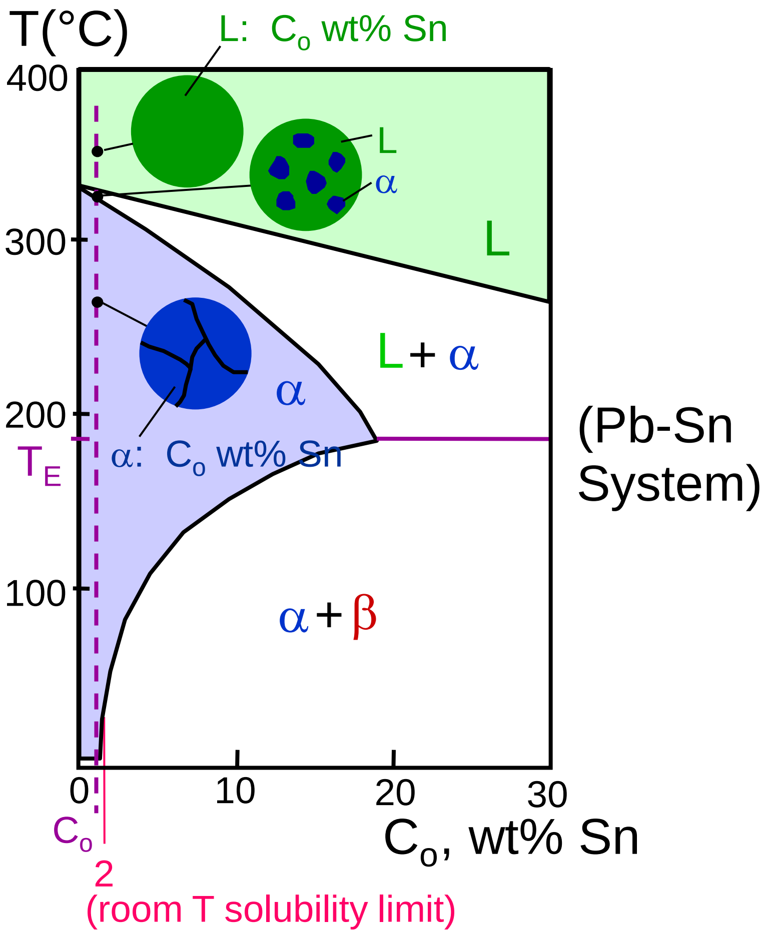
Adapted from Fig. 9.8, Callister 7e.

Microstructures in Eutectic Systems: I

- C_o < 2 wt% Sn
- Result:

  --at extreme ends

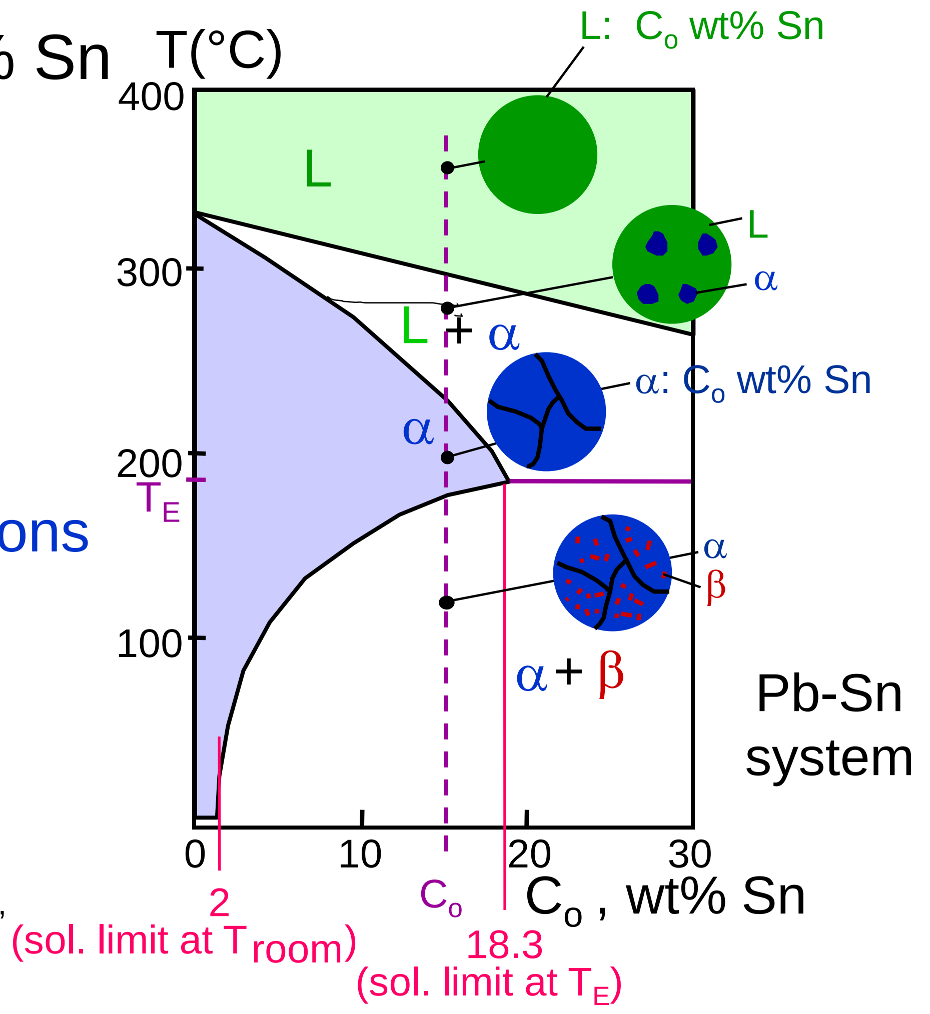
  --polycrystal of α grains
  i.e., only one solid phase.



Adapted from Fig. 9.11, Callister 7e.

Microstructures in Eutectic Systems: II

- 2 wt% Sn < C_o < 18.3 wt% Sn
- Result:
  - Initially liquid + α
  - then α alone
  - finally two phases
    - α polycrystal
    - fine β-phase inclusions

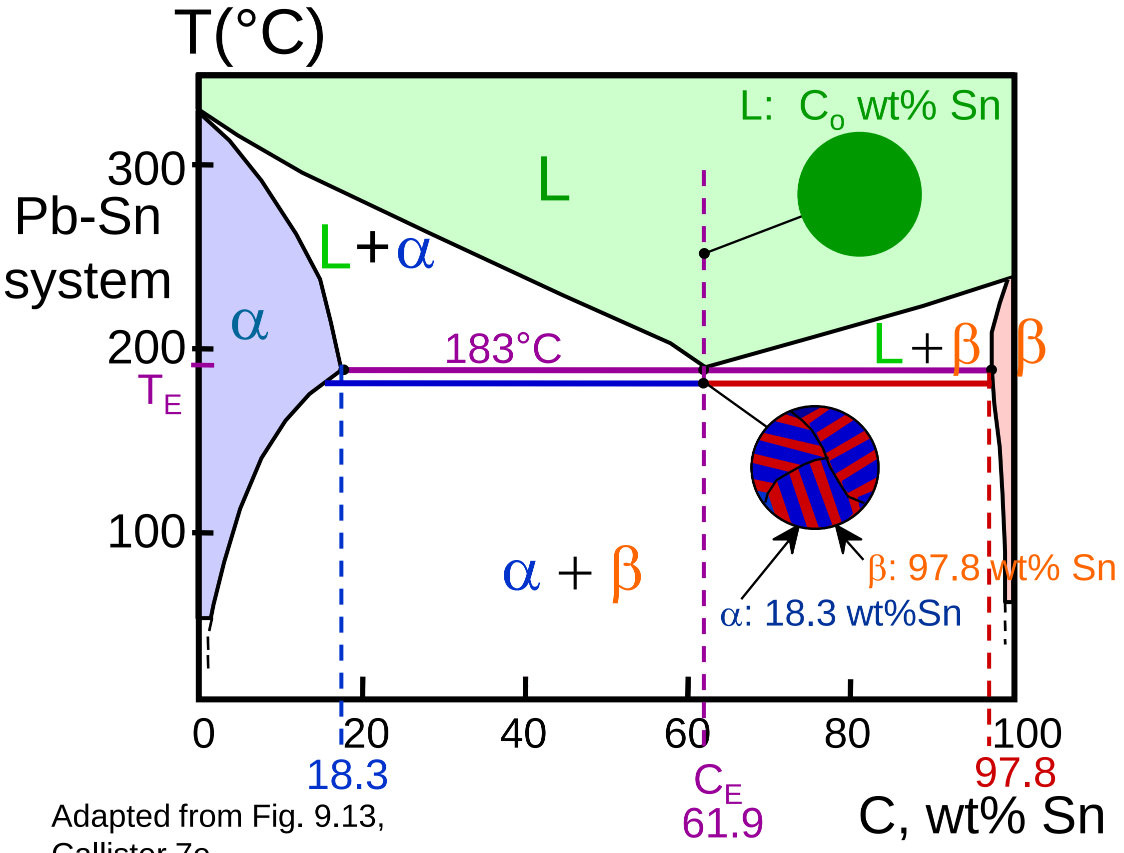


Pb-Sn system

Adapted from Fig. 9.12, Callister 7e.

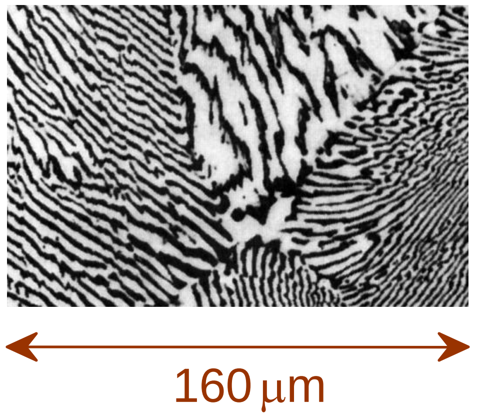
Microstructures in Eutectic Systems: III

- $C_o = C_E$
- Result: Eutectic microstructure (lamellar structure)
  --alternating layers (lamellae) of α and β crystals.



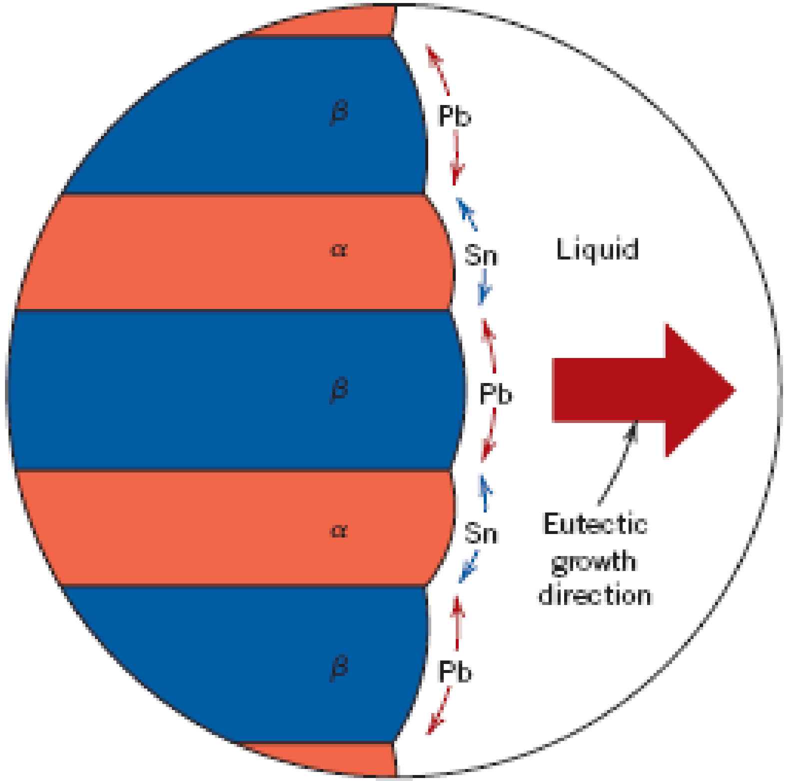
Adapted from Fig. 9.13, Callister 7e.

Micrograph of Pb-Sn eutectic microstructure

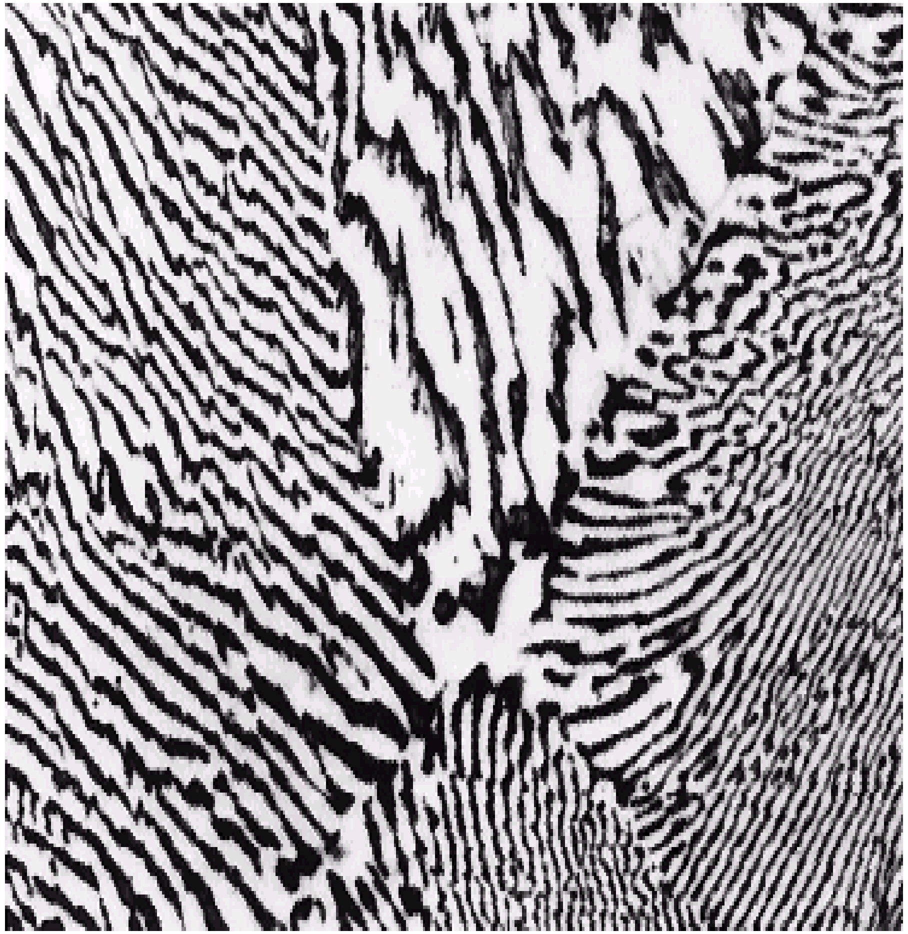


Adapted from Fig. 9.14, Callister 7e.

Lamellar Eutectic Structure

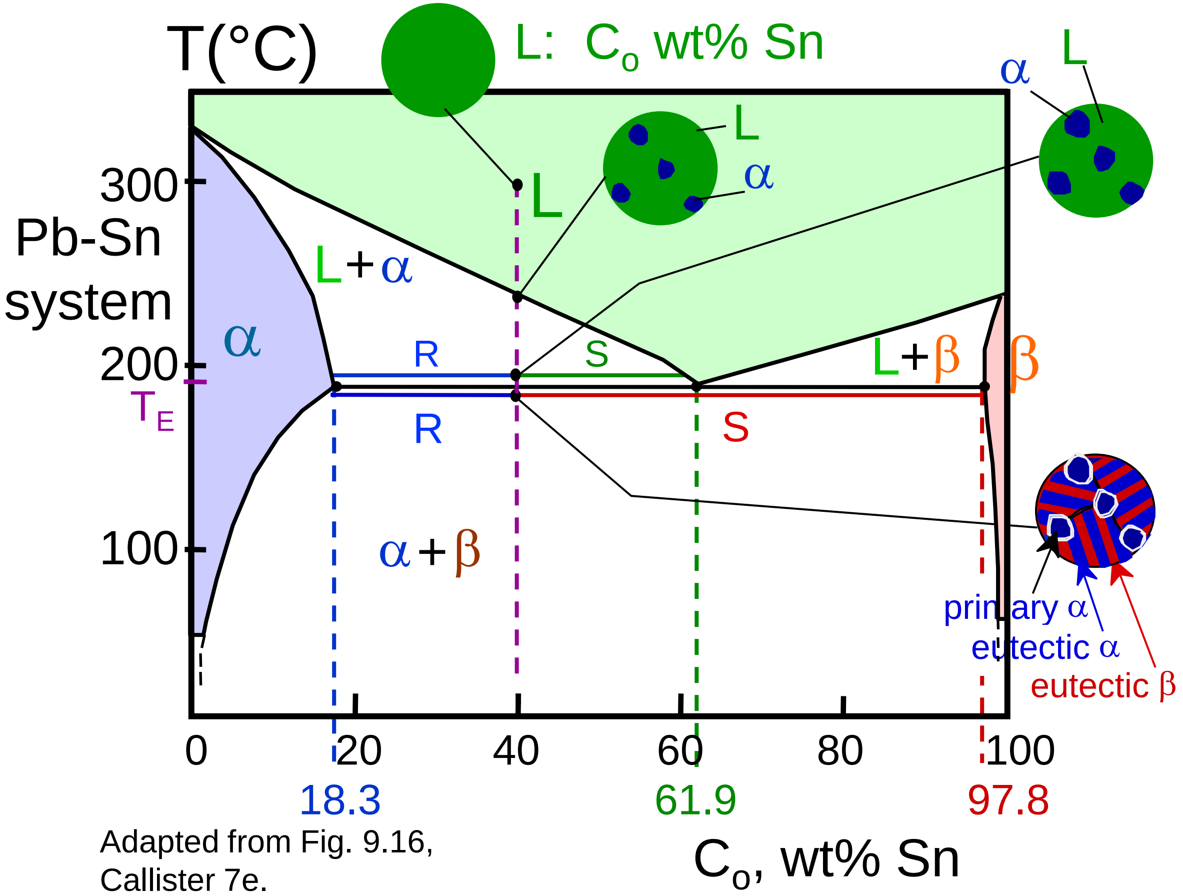


Adapted from Figs. 9.14 & 9.15, Callister 7e.



Microstructures in Eutectic Systems: IV

- 18.3 wt% Sn < C_o < 61.9 wt% Sn
- Result: α crystals and a eutectic microstructure



Adapted from Fig. 9.16, Callister 7e.

- Just above T_E :

C_α = 18.3 wt% Sn

C_L = 61.9 wt% Sn

$$W_\alpha = \frac{S}{R+S} = 50 \text{ wt\%}$$

W_L = (1- W_α) = 50 wt%

- Just below T_E :

C_α = 18.3 wt% Sn

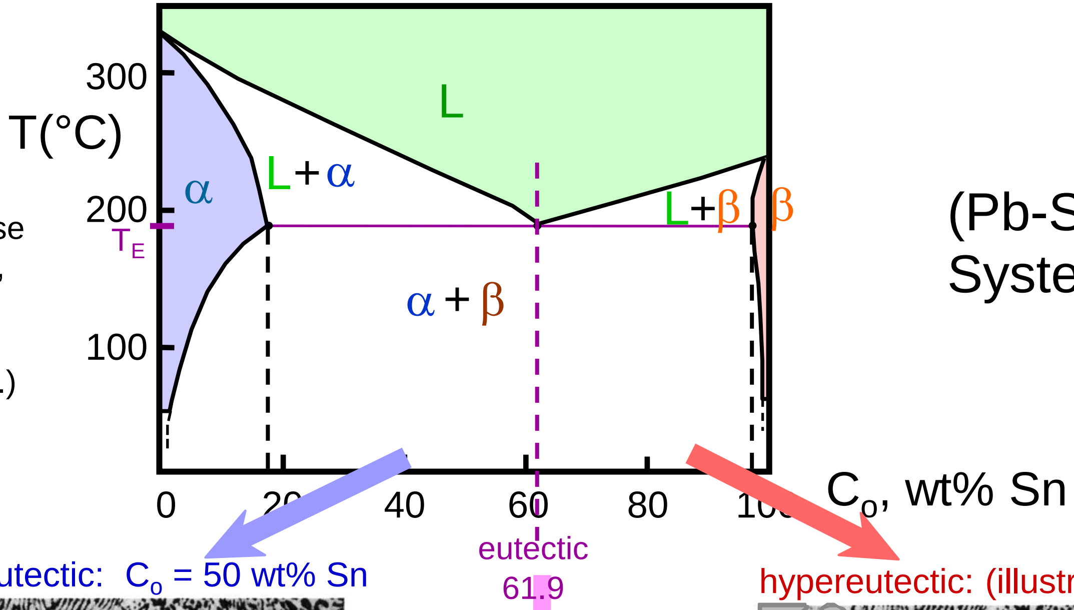
C_β = 97.8 wt% Sn

$$W_\alpha = \frac{S}{R+S} = 73 \text{ wt\%}$$

W_β = 27 wt%

Hypoeutectic & Hypereutectic

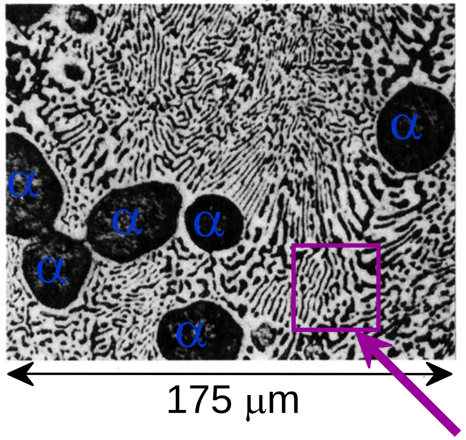
Adapted from Fig. 9.8, Callister 7e. (Fig. 9.8 adapted from Binary Phase Diagrams, 2nd ed., Vol. 3, T.B. Massalski (Editor-in-Chief), ASM International, Materials Park, OH, 1990.)



(Pb-Sn System)

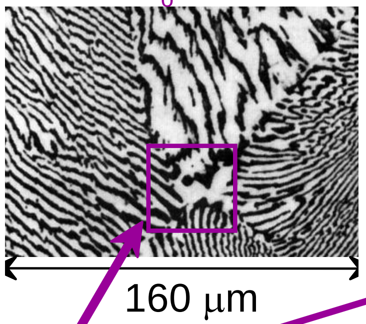
(Figs. 9.14 and 9.17 from Metals Handbook, 9th ed., Vol. 9, Metallography and Microstructures, American Society for Metals, Materials Park, OH, 1985.)

hypoeutectic: C_o = 50 wt% Sn



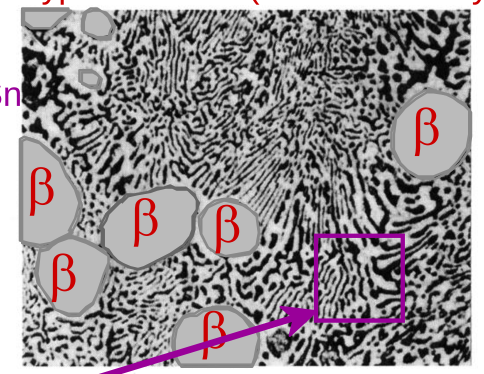
Adapted from Fig. 9.17, Callister 7e.

eutectic: C_o = 61.9 wt% Sn



Adapted from Fig. 9.14, Callister 7e.

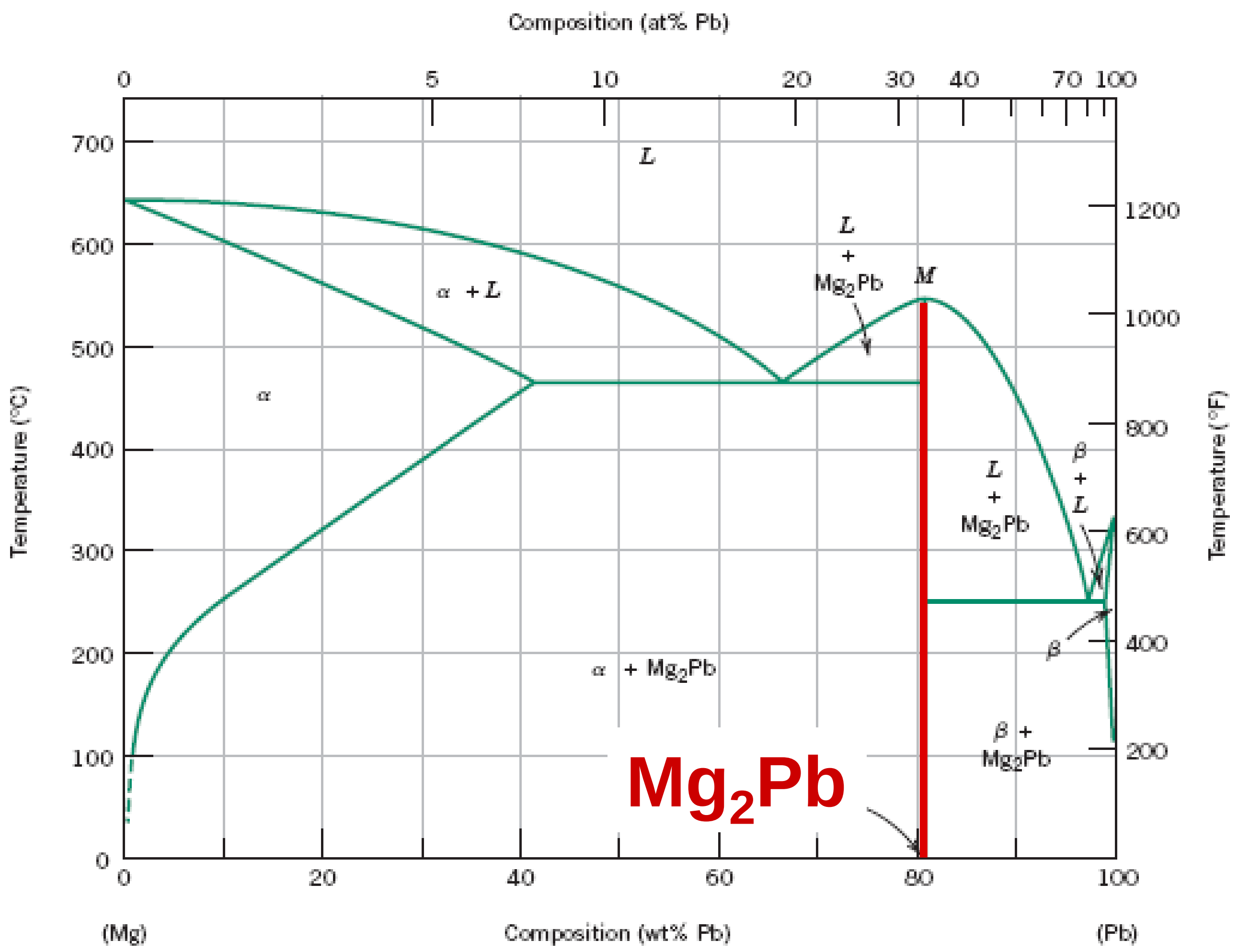
hypereutectic: (illustration only)



Adapted from Fig. 9.17, Callister 7e. (Illustration only)

eutectic micro-constituent

Intermetallic Compounds



Adapted from
Fig. 9.20, Callister 7e.

Note: intermetallic compound forms a line - not an area - because stoichiometry (i.e. composition) is exact.

Eutectoid & Peritectic

- Eutectic - liquid in equilibrium with two solids

$$L \underset{\text{heat}}{\overset{\text{cool}}{\rightleftharpoons}} \alpha + \beta$$

- Eutectoid - solid phase in equation with two solid phases

$$S_2 \rightleftharpoons S_1 + S_3$$

intermetallic compound
- cementite

$$\gamma \underset{\text{heat}}{\overset{\text{cool}}{\rightleftharpoons}} \alpha + Fe_3C \qquad (727^oC)$$

- Peritectic - liquid + solid 1 → solid 2 (Fig 9.21)

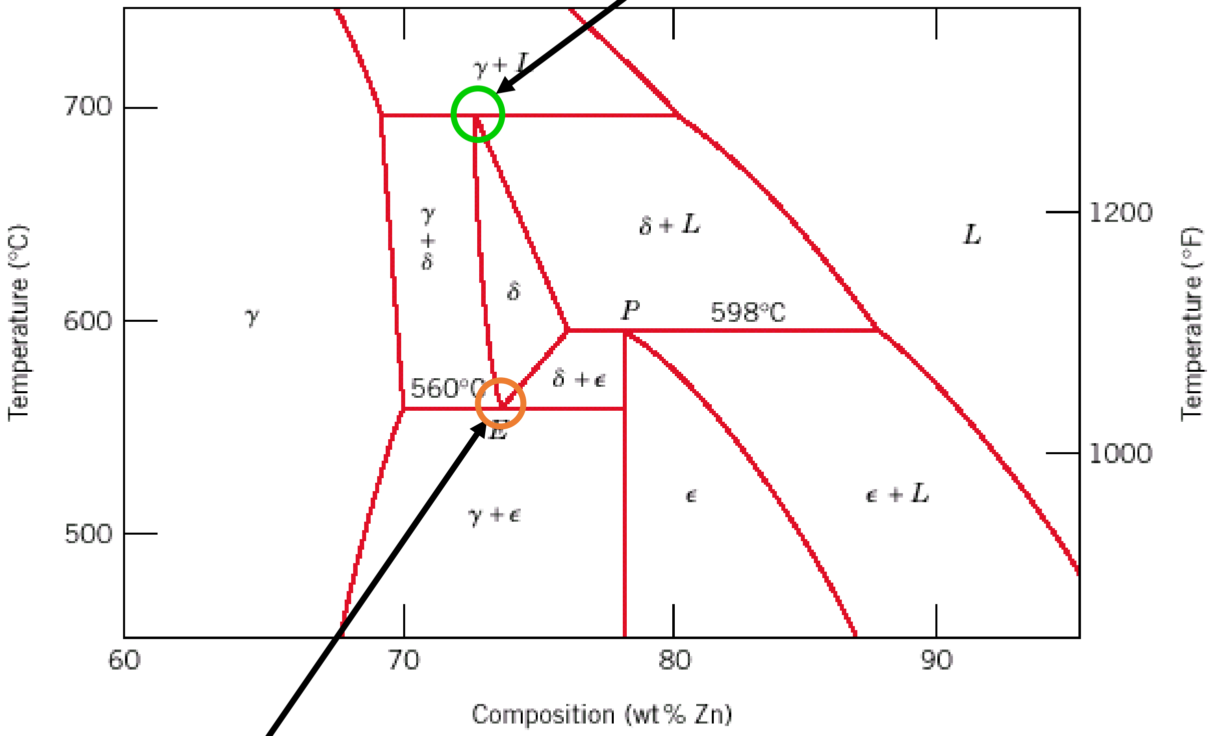
$$S_1 + L \rightleftharpoons S_2$$

$$\delta + L \underset{\text{heat}}{\overset{\text{cool}}{\rightleftharpoons}} \gamma \qquad (1493^oC)$$

Eutectoid & Peritectic

Peritectic transition $\gamma + L \rightleftharpoons \delta$

Cu-Zn Phase diagram



Adapted from
Fig. 9.21, Callister 7e.

Eutectoid transition $\delta \rightleftharpoons \gamma + \varepsilon$

Iron-Carbon (Fe-C) Phase Diagram

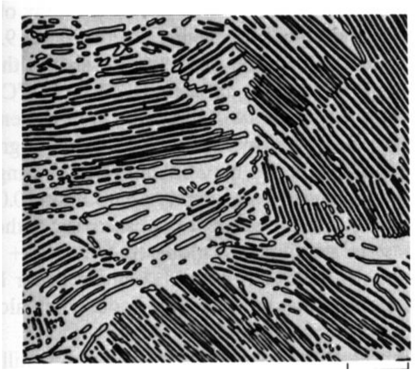
- 2 important points

-Eutectic (A):

$L \Rightarrow \gamma + Fe_3C$

-Eutectoid (B):

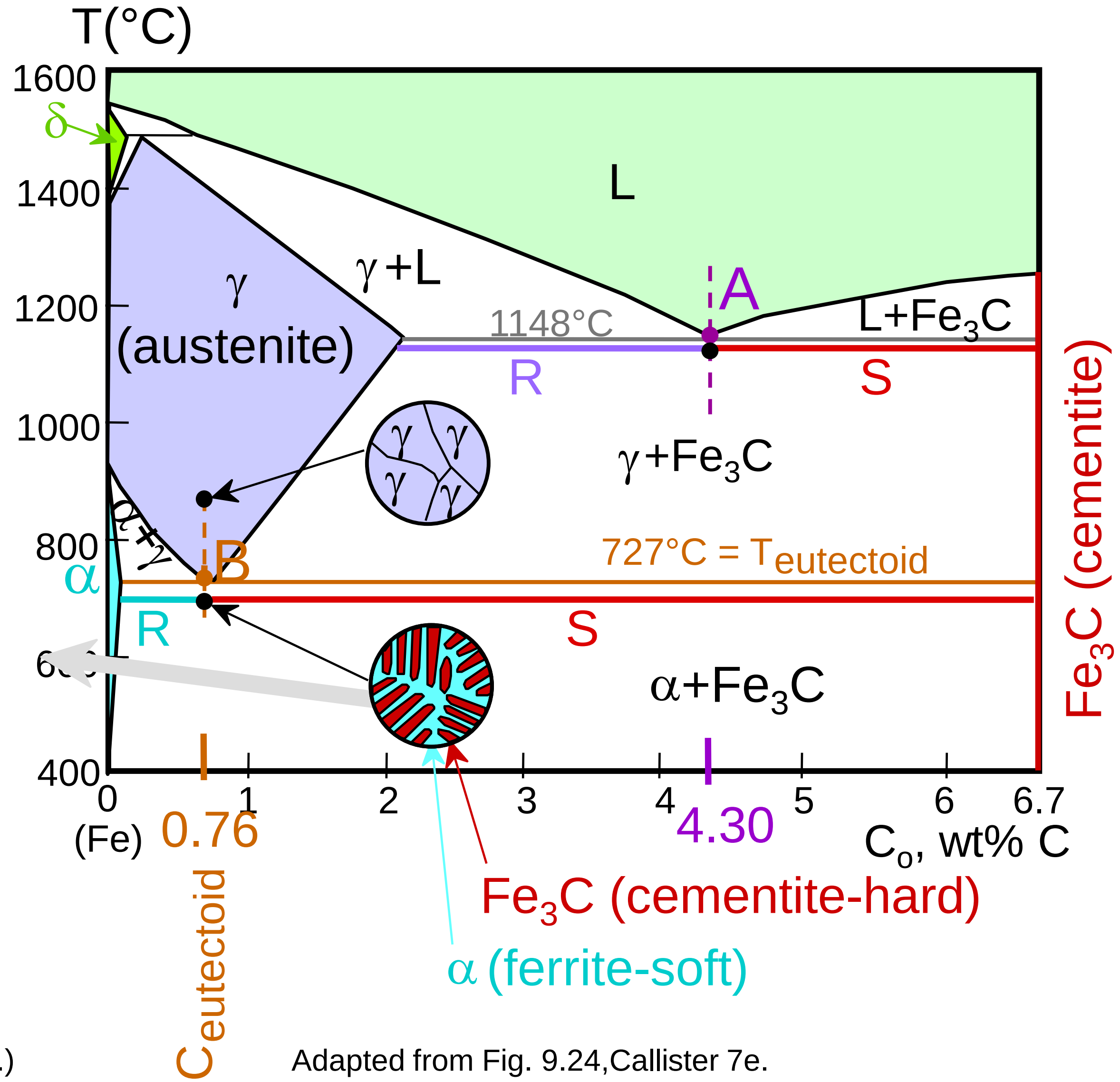
$\gamma \Rightarrow \alpha + Fe_3C$



120 μm

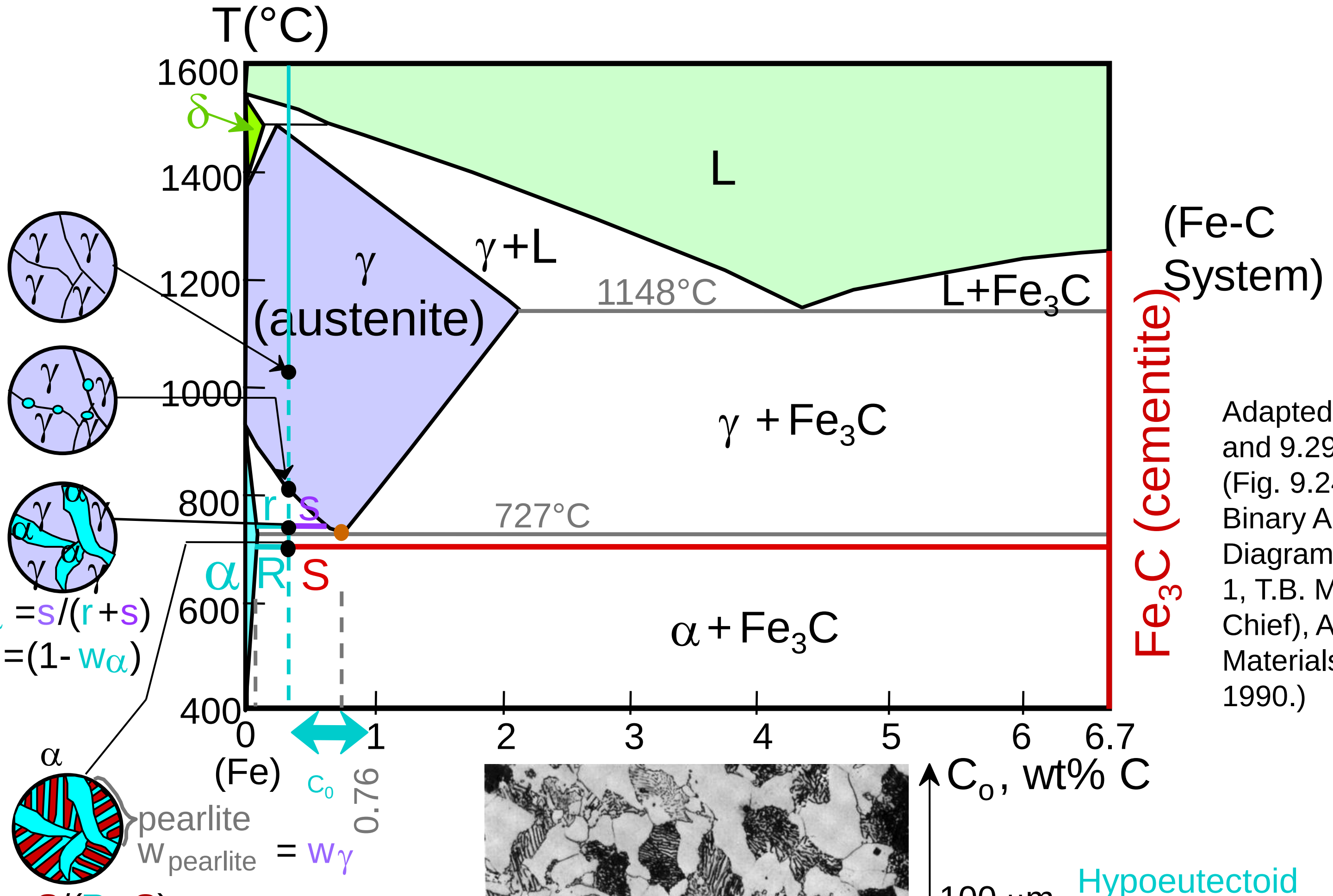
Result: Pearlite = alternating layers of α and Fe_3C phases

(Adapted from Fig. 9.27, Callister 7e.)



Adapted from Fig. 9.24,Callister 7e.

Hypoeutectoid Steel



$w_\alpha = s/(r+s)$

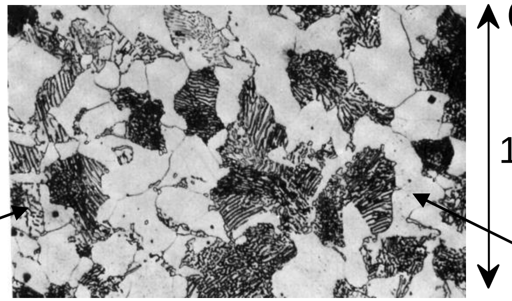
$w_\gamma = (1 - w_\alpha)$

$w_{pearlite} = w_\gamma$

$w_\alpha = S/(R+S)$

$w_{Fe_3C} = (1 - w_\alpha)$

Adapted from Figs. 9.24 and 9.29, Callister 7e. (Fig. 9.24 adapted from Binary Alloy Phase Diagrams, 2nd ed., Vol. 1, T.B. Massalski (Ed.-in-Chief), ASM International, Materials Park, OH, 1990.)



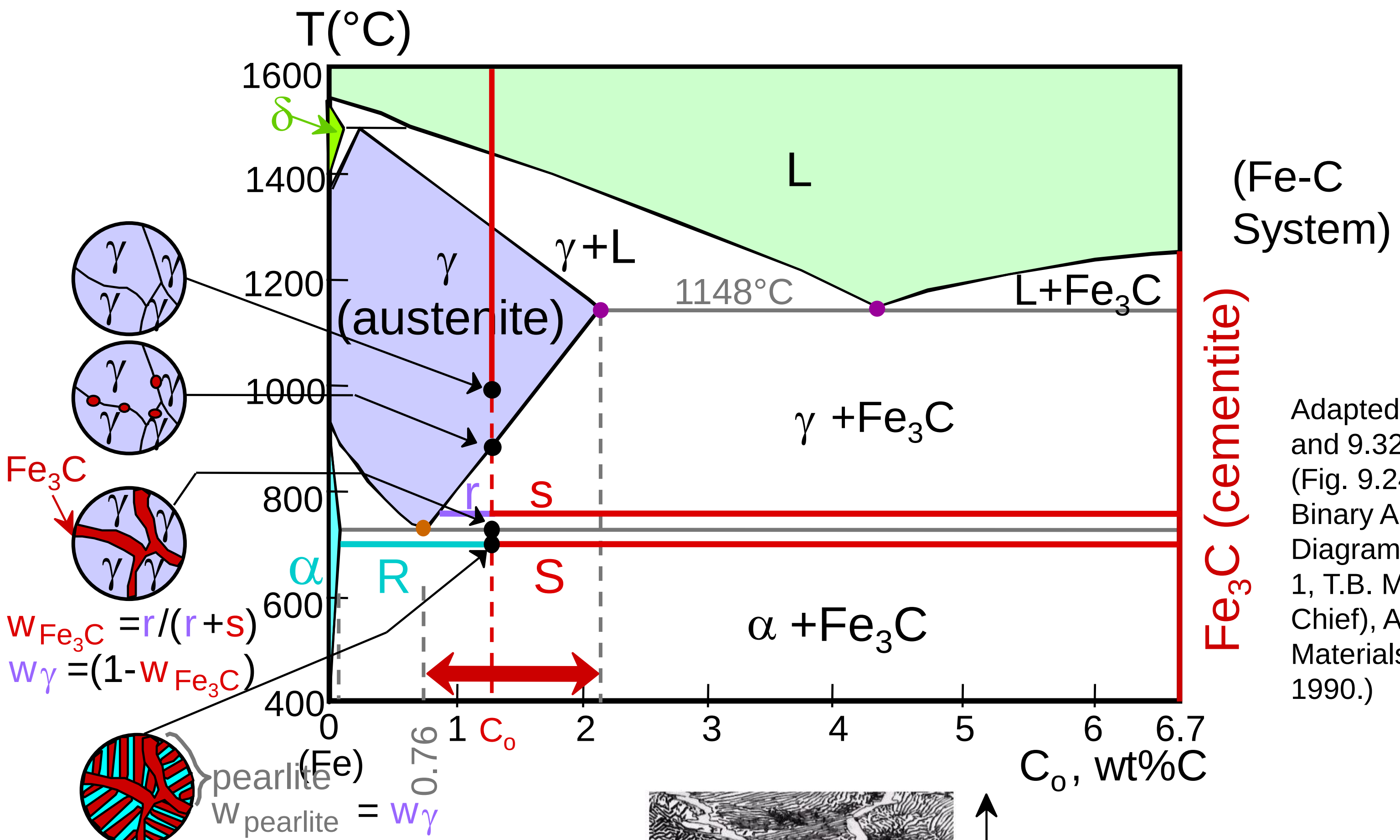
pearlite

proeutectoid ferrite

Hypoeutectoid steel

Adapted from Fig. 9.30, Callister 7e.

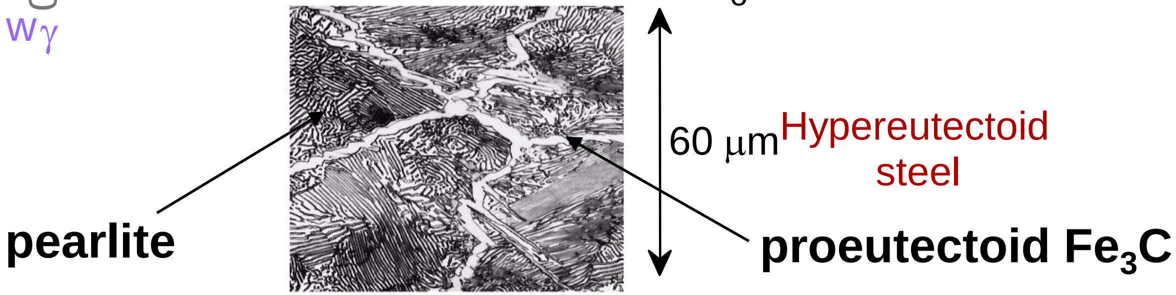
Hypereutectoid Steel



$w_\alpha = S/(R+S)$

$w_{Fe_3C} = (1 - w_\alpha)$

Adapted from Figs. 9.24 and 9.32,Callister 7e. (Fig. 9.24 adapted from Binary Alloy Phase Diagrams, 2nd ed., Vol. 1, T.B. Massalski (Ed.-in-Chief), ASM International, Materials Park, OH, 1990.)



Adapted from Fig. 9.33,Callister 7e.

Example: Phase Equilibria

For a 99.6 wt% Fe-0.40 wt% C at a temperature just below the eutectoid, determine the following

a) composition of Fe_3C and ferrite (α)
b) the amount of carbide (cementite) in grams that forms per 100 g of steel
c) the amount of pearlite and proeutectoid ferrite (α)

Chapter 9 – Phase Equilibria

Solution: a) composition of Fe_3C and ferrite (α)

b) the amount of carbide (cementite) in grams that forms per 100 g of steel

C_O = 0.40 wt% C

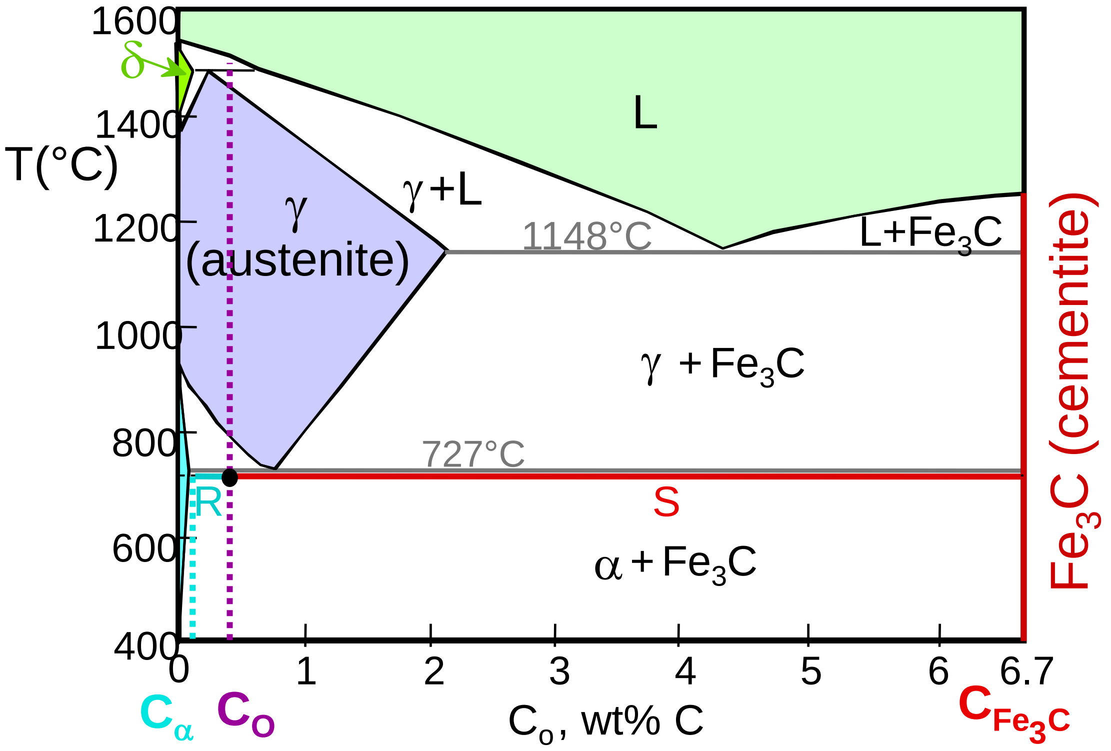
C_α = 0.022 wt% C

C_{Fe_3C} = 6.70 wt% C

$$\frac{Fe_3C}{Fe_3C+\alpha} = \frac{C_o - C_\alpha}{C_{Fe_3C} - C_\alpha} \text{x}100$$

$$= \frac{0.4-0.022}{6.7-0.022} x\,100 = 5.7\text{g}$$

$$\boxed{\begin{array}{l} Fe_3C = 5.7\text{ g} \\ \alpha = 94.3\text{ g} \end{array}}$$



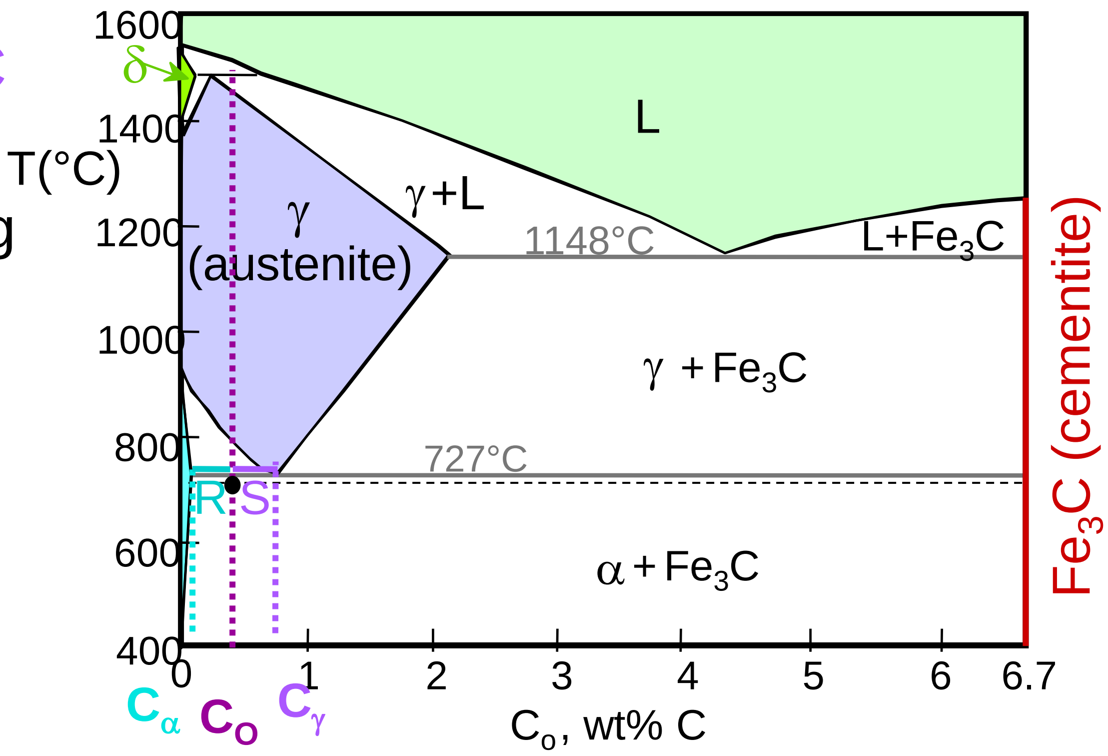
Chapter 9 – Phase Equilibria

c. the amount of pearlite and proeutectoid ferrite (α)

note: amount of pearlite = amount of γ just above T_E

C_o = 0.40 wt% C
C_α = 0.022 wt% C
$C_{pearlite}$ = C_γ = 0.76 wt% C

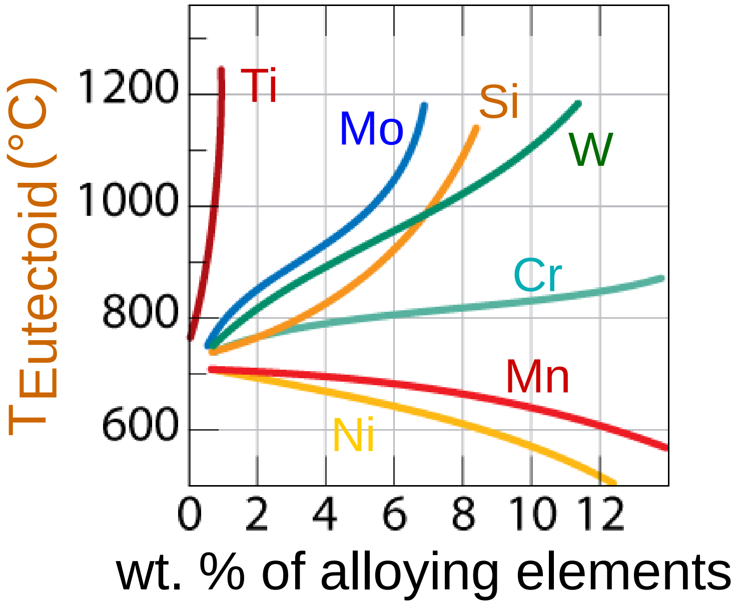
$$\frac{\gamma}{\gamma+\alpha} = \frac{C_o - C_\alpha}{C_\gamma - C_\alpha} \text{ x } 100 = 51.2 \text{ g}$$



pearlite = 51.2 g
proeutectoid α = 48.8 g

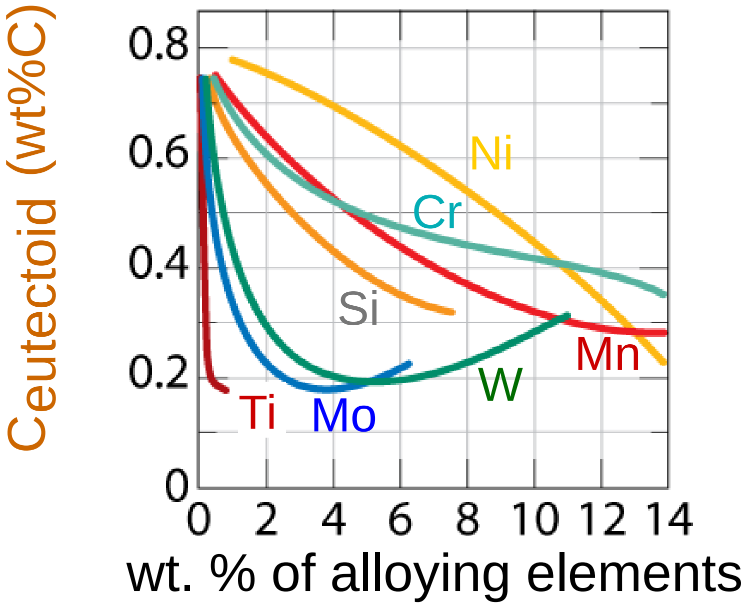
Alloying Steel with More Elements

- $T_{eutectoid}$ changes:



Adapted from Fig. 9.34,Callister 7e. (Fig. 9.34 from Edgar C. Bain, Functions of the Alloying Elements in Steel, American Society for Metals, 1939, p. 127.)

- $C_{eutectoid}$ changes:



Adapted from Fig. 9.35,Callister 7e. (Fig. 9.35 from Edgar C. Bain, Functions of the Alloying Elements in Steel, American Society for Metals, 1939, p. 127.)

Summary

- Phase diagrams are useful tools to determine:

  --the number and types of phases,
  --the wt% of each phase,
  --and the composition of each phase

  for a given T and composition of the system.

- Alloying to produce a solid solution usually

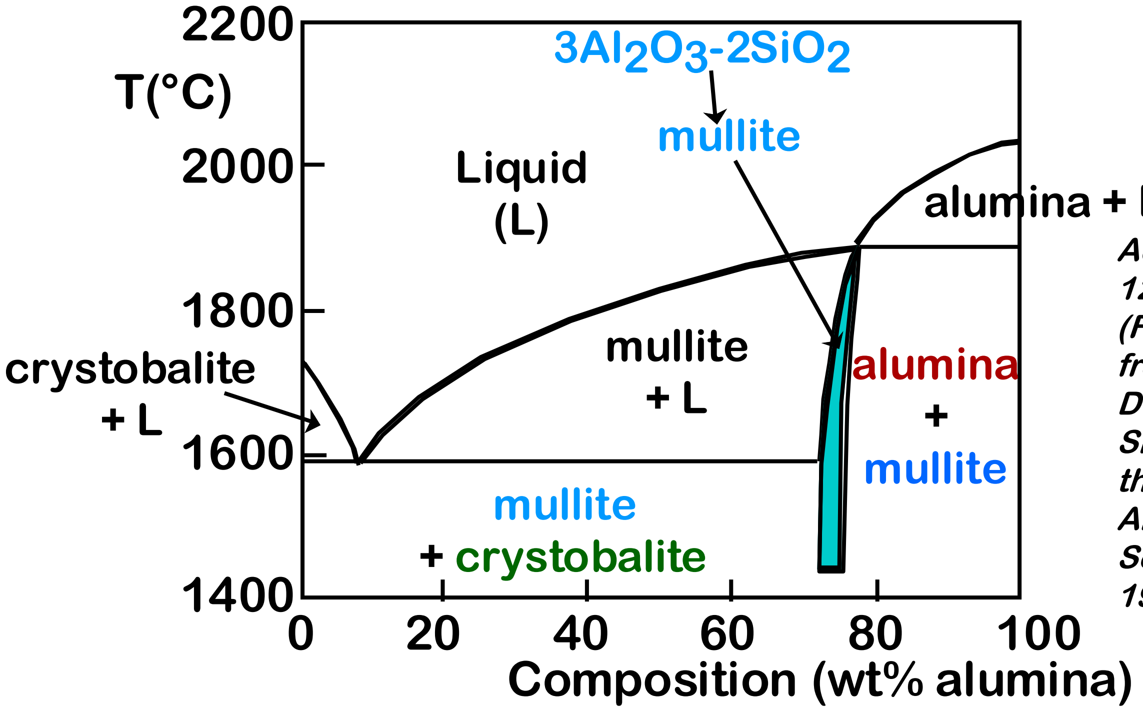
  --increases the tensile strength (TS)
  --decreases the ductility.

- Binary eutectics and binary eutectoids allow for a range of microstructures.

APPLICATION: REFRACTORIES

- ***Need a material to use in high temperature furnaces.***
- ***Consider Silica (SiO_2) - Alumina (Al_2O_3) system.***
- ***Phase diagram shows:***

mullite, alumina, and crystobalite (made up of SiO_2) tetrahedra as candidate refractories.



Adapted from Fig. 12.27, Callister 6e. (Fig. 12.27 is adapted from F.J. Klug and R.H. Doremus, "Alumina Silica Phase Diagram in the Mullite Region", J. American Ceramic Society 70(10), p. 758, 1987.)

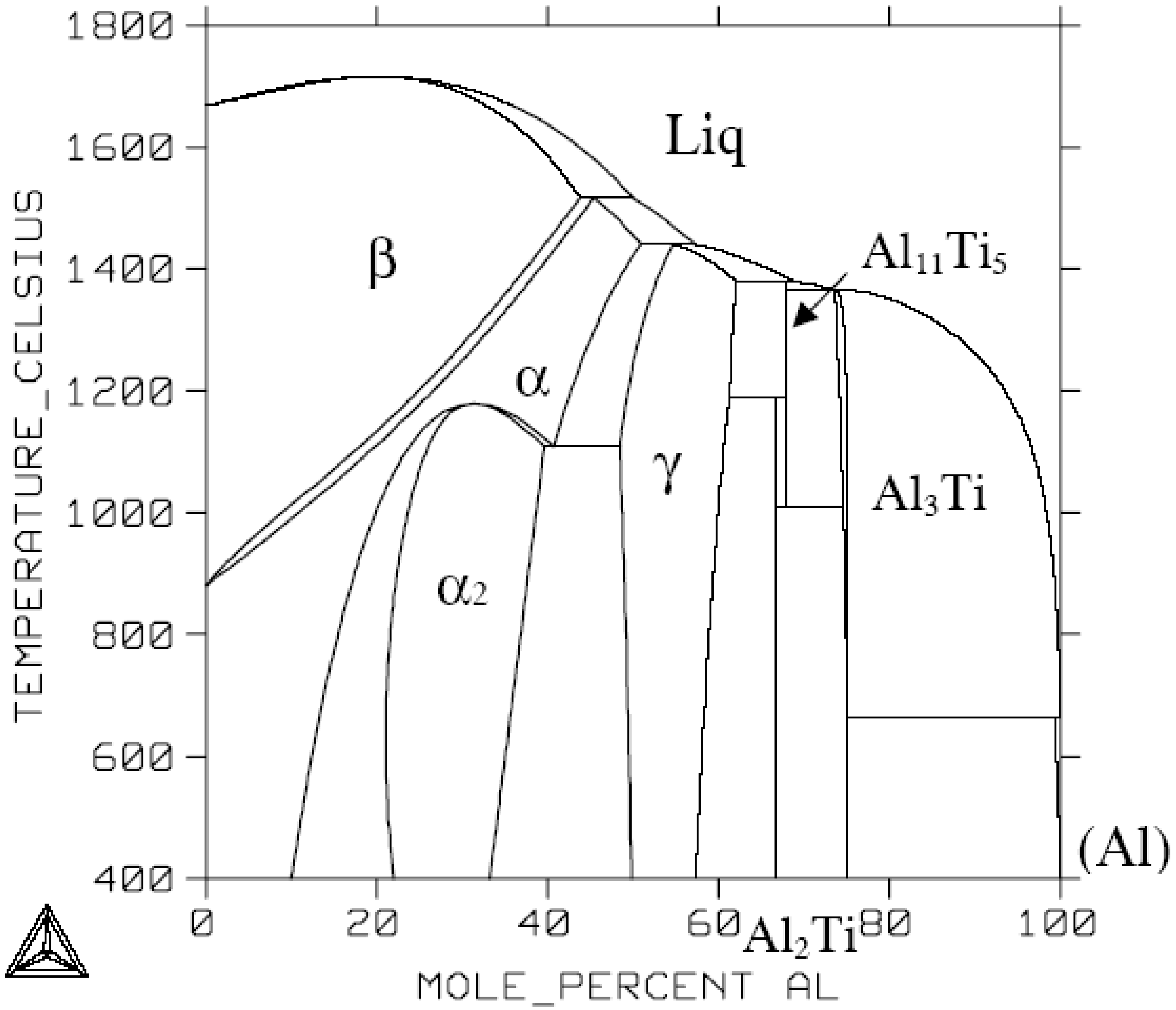
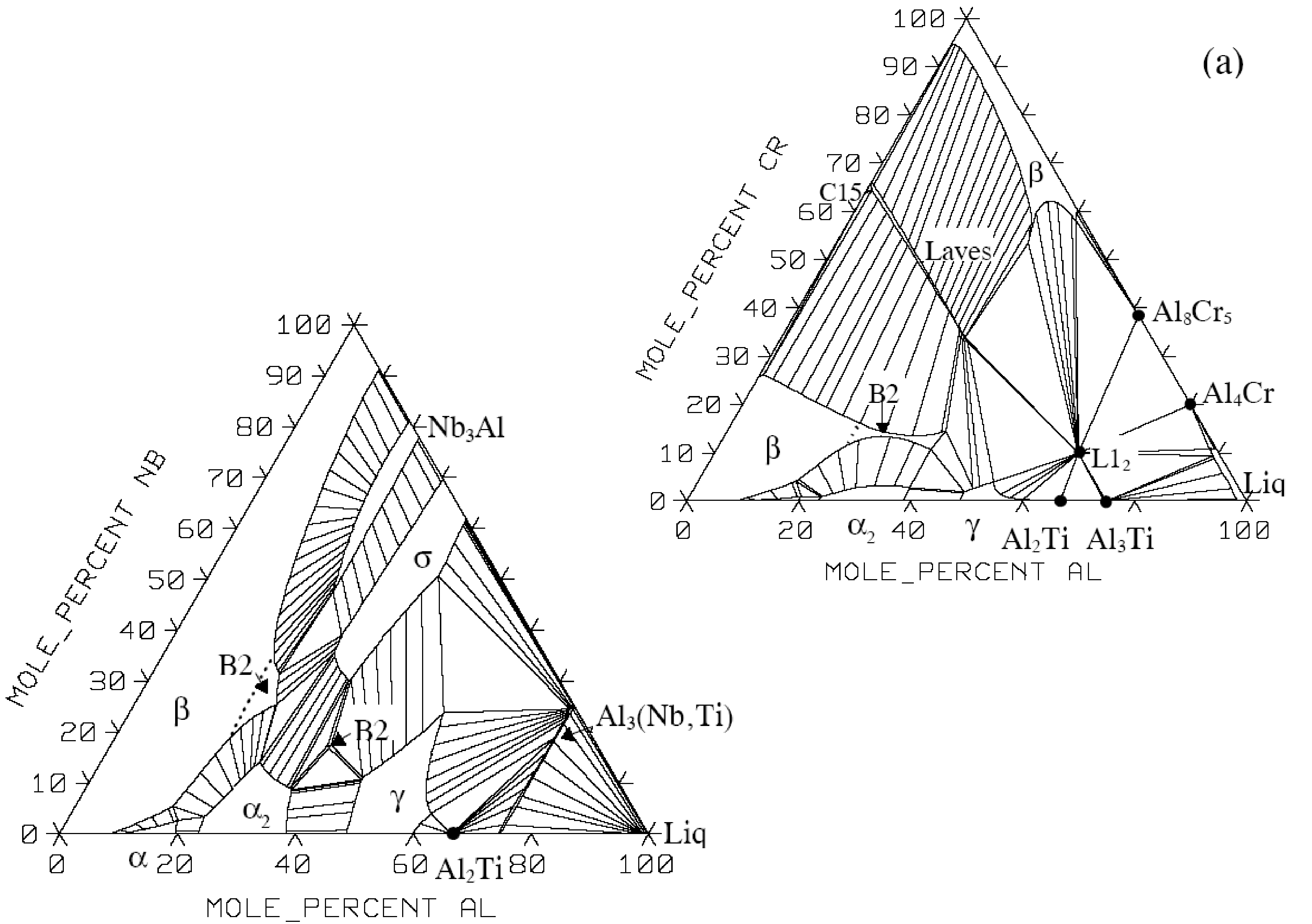


Fig.6 Calculated Ti-Al Phase Diagram[28]


(a)
MOLE_PERCENT NB
MOLE_PERCENT AL
Nb$_3$Al
σ
β
B2
B2
α$_2$
γ
α
Al$_3$(Nb,Ti)
Liq
Al$_2$Ti
MOLE_PERCENT CR
MOLE_PERCENT AL
C15
Laves
β
B2
β
α$_2$
γ
Al$_8$Cr$_5$
Al$_4$Cr
L1$_2$
Liq
Al$_2$Ti
Al$_3$Ti